CSR: Physically conditioned scale-transfer super-resolution across optical missions

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Abstract

Optical Earth observation missions differ in bandpass and native spatial sampling, limiting the reuse of sensor-specific super-resolution models. We introduce CSR, a physically conditioned scale-transfer model for variable band sets and requested output grids. Multi-mission native-resolution Wald pretraining shares a spatial-detail prior without paired cross-sensor supervision. Wavelength, bandwidth and effective pixel spacing condition a shared spatial–spectral encoder and coordinate renderer; MTF-guided composition anchors reconstruction to the observations. CSR achieves the largest mean OpenSR structural improvement on all four scale-four suites and lies on the observed PSNR–improvement frontier in 4 of five suites. BreizhSR adaptation gains 0.46 dB corrected PSNR over bicubic and 0.41 dB over random initialization, with the highest structural improvement among matched methods. On 174 held-out PROBA-V scenes, validation-selected adaptation improves corrected PSNR over bicubic by 0.58 dB in RED and 0.47 dB in NIR, and over random initialization by 0.45 and 0.37 dB. CSR supplies a reusable multi-mission prior for spatial reconstruction and mission adaptation.

Keywords: Optical remote sensing, Super-resolution, Physical conditioning, Multi-mission learning, Wald supervision

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1. Introduction

Optical satellite archives observe related land surfaces through different spectral bands, spatial response functions and sampling grids. Yet super-resolution models commonly tie their input channels and reconstruction heads to a particular sensor and magnification. Deploying them across an archive consequently requires maintaining separate models for different missions, band configurations or output scales. This fragmentation increases training and deployment costs and makes fine-resolution spatial information difficult to share across missions.

Existing approaches address parts of this problem. MDSR shares a feature extractor but retains distinct preprocessing and upsampling modules for selected integer factors (Lim et al., 2017). Overscaling and coordinate representations instead support variable output sampling (Behjati et al., 2021; Chen et al., 2021). Sensor-specific guided models exploit complementary bands but retain fixed input and target-band configurations (Lanaras et al., 2018). Scale flexibility alone therefore leaves the heterogeneous-band problem unresolved. A common Earth-observation reconstruction model should accept the available optical bands, understand their physical sampling and spectral properties, and render them on the grid required by the application.

Recent remote-sensing super-resolution research emphasizes the gap between simulated degradations and real sensor observations, alongside the need for degradation-aware models and broader evaluation (Qi et al., 2026). Coordinate representations offer a complementary approach to variable output sampling (Chen et al., 2021). LMF separates latent feature decoding from low-dimensional coordinate rendering (He and Jin, 2024), while DOFA motivates wavelength-conditioned processing for heterogeneous sensor inputs (Xiong et al., 2024). SEN2SR emphasizes radiometric and spatial consistency in Sentinel-2 reconstruction (Aybar et al., 2026). These directions suggest a reusable reconstruction backbone whose processing is driven by band physics rather than a fixed mission-specific channel head. Assessing such a model requires both native-band fidelity and the introduced or omitted structure in finer reconstructions. OpenSR-test provides such a decomposition alongside conventional fidelity measures (Aybar et al., 2024).

Cross-sensor learning can also use a fine-resolution source sensor and target-specific degradation models: Qiu et al. (2023) combine a target-sensor degradation pool with edge-guided reconstruction. CSR adopts a shared multi-mission interface conditioned by band physics.

We develop a single physically conditioned reconstruction model across optical missions. The approach combines per-band feature modulation, interleaved spatial–spectral attention and a coordinate-conditioned residual decoder. An MTF-guided frequency composition limits the low-frequency contribution of the learned residual. Pretraining derives supervision from each mission's native observations using a continuous-scale Wald protocol. Real-reference supervision is introduced separately in the task-specific adaptation experiments.

We call the shared learning principle *scale-transfer*: observations from missions with fine native ground sampling distance contribute a detail prior that the same parameters reuse when reconstructing coarser imagery. The training loss still uses within-mission Wald targets. The additional assumption is cross-mission reuse of this prior through shared processing and continuous physical descriptors, beyond a sensor-specific scale-invariance model.

Our contribution is a common reconstruction interface for heterogeneous optical bands and continuously specified output sampling, trained from native observations rather than scarce cross-sensor pairs. We evaluate its structural recovery on OpenSR, paired cross-sensor adaptation on BreizhSR and adaptation to a new mission using PROBA-V. The experiments and mission illustrations connect the model's shared parameterization to the different reconstruction tasks it can serve.

2. Method

CSR reconstructs optical bands on a requested finer grid while using each band's wavelength, bandwidth and pixel spacing to guide processing. Figure 1 follows the image through five operations: describe the input bands, extract local patterns, exchange information across locations and bands, predict detail at requested pixels, and combine that detail with the observed image. Native observations also provide the pretraining targets: an image is blurred and resampled, then reconstructed at its original sampling.

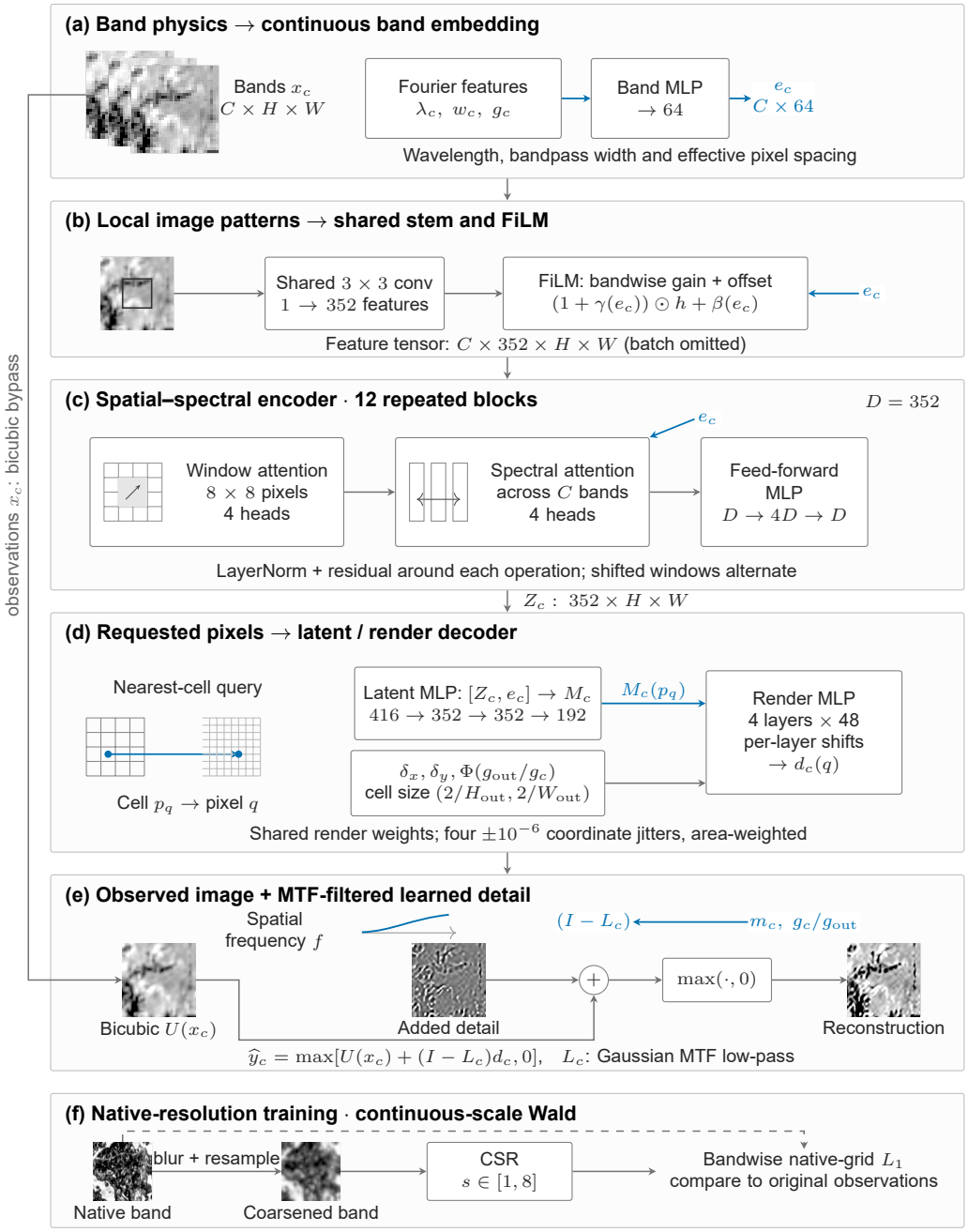


Figure 1: **CSR couples image content and band physics.** Image and grid examples accompany the modules and tensor dimensions. Blue marks conditioning and modulation; the bottom strip shows native-grid pretraining.

The visible-band thumbnails illustrate a recorded OpenSR example; added detail shows the final CSR-minus-bicubic difference. Training thumbnails illustrate a recorded native-band example. Feature, response and grid icons are schematic.

2.1. Representing bands and their physical properties

The input $x \in \mathbb{R}^{C\times H\times W}$ contains a variable set of bands resampled onto a common working grid. Band c has centre wavelength λ_c, spectral bandwidth w_c, effective input ground sampling distance g_c, and an assumed MTF at Nyquist m_c. The requested output sampling distance is g_{out}. Instrument centre wavelengths and bandwidths provide a compact approximation to spectral response.

Sinusoidal features of $\lambda_c/1000$, $w_c/1000$ and $g_c/60$ pass through an MLP to produce a band embedding e_c. A shared 3×3 filter extracts local patterns from each band. The band descriptor adjusts their gain and offset, allowing the same filters to respond differently to different spectral and spatial measurements (FiLM modulation; (Perez et al., 2018)):

$$z_c = (1 + \gamma(e_c)) \odot h(x_c) + \beta(e_c). \tag{1}$$

This feature extractor shares parameters across channels while allowing their physical descriptors to affect processing. No learned mission identity token is enabled. Sentinel-2, Landsat and VENuS inputs use their recorded reflectance conventions; NAIP uses scaled uncalibrated image DN. The latter is not a physical reflectance measurement. Pretraining retains each mission's recorded radiometry; the BreizhSR adaptation experiment uses reference-side harmonization as described in Section 3.4.

2.2. Sharing information across pixels and bands

Two complementary operations combine the extracted patterns. Spatial attention exchanges information among nearby pixels; spectral attention exchanges information across the supplied bands. Each encoder block applies these operations followed by a feed-forward network, with normalization and residual connections. Shifted spatial windows alternate across blocks. Band positional features enter the spectral computation. CSR uses twelve blocks, feature width 352, four attention heads, eight-pixel windows and a 64-dimensional band embedding. The attention operates on the supplied channel set, enabling different numbers of input bands without a fixed output-channel head.

2.3. Rendering pixels on the requested grid

To predict a new pixel, CSR combines the encoded image content, the band descriptor, the pixel position and the requested sampling ratio. Following the latent/render decomposition of LMF (He and Jin, 2024), an MLP with hidden width 352 combines z_c and e_c into per-cell features that adjust the shared render function. The render MLP has four layers of width 48. For an output query q, its inputs are the position relative to the selected latent-cell centre, sinusoidal features of g_{out}/g_c, and the output-cell dimensions. Per-layer shifts derived from the latent modulation carry the image content. Output queries are evaluated in chunks to bound inference memory.

The implemented query rule selects the nearest latent cell and applies four coordinate perturbations of magnitude 10^{-6} with opposite-area weighting. Away from latent-cell boundaries these perturbations usually select the same cell. The render function uses relative coordinates within each selected cell; cell transitions can therefore introduce discontinuities. We evaluate full four-neighbour queries separately as an inference diagnostic.

The latent and render mappings can be written as

$$\begin{aligned} M_c(p) &= f_\theta([z_c(p), e_c]), \\ d_c(q) &= R_\theta([\delta_x, \delta_y, \Phi(g_{\text{out}}/g_c), 2/H_{\text{out}}, 2/W_{\text{out}}]; M_c(p_q)) \,. \end{aligned} \tag{2}$$

where p_q is the selected latent-cell centre and Φ is the inverse-scale Fourier encoding. Equation (2) shows one query; the implemented jitter predictions are area-weighted before residual composition. The modulation contains one width-48 shift vector for each render layer.

The decoder predicts a residual $d_c(q)$ over bicubic interpolation:

$$y_c^{\text{raw}}(q) = U(x_c)(q) + d_c(q). \tag{3}$$

CSR contains 24,572,065 parameters and is pretrained for 500000 optimizer updates.

2.4. MTF-guided frequency composition

Motivated by consistency-aware Sentinel-2 reconstruction (Aybar et al., 2026), we compose frequency components using bandwise physical descriptors. For Gaussian spatial response, an MTF m_c at native Nyquist gives

$$\sigma_c = \frac{\sqrt{-2\log m_c}}{\pi}, \qquad H_c(f) = \exp\left[-2\pi^2(\sigma_c g_c/g_{\text{out}})^2\|f\|^2\right], \tag{4}$$

where f is measured in cycles per output pixel. Let L_c apply this mask in the Fourier domain. The output is

$$\widehat{y}_c = \max(y_c^{\text{raw}} - L_c(y_c^{\text{raw}}) + L_c(U(x_c)), 0)\,. \tag{5}$$

Equation (5) anchors low-frequency content to the observation while preserving learned detail above the band response. Its smooth mask and clipping give approximate consistency, assessed through low-frequency agreement and degradation-back error.

2.5. Native-resolution Wald supervision

Training samples a degradation factor $s \in [1, 8]$. Each native band is blurred using its assumed MTF and resampled to a common coarser working grid. Effective conditioning changes to $g_c^{\text{input}} = s g_c^{\text{native}}$. The working grid remains 64×64, while the centred or randomly selected native target crop grows with s. Coarser native bands are represented on the shared finest grid; recovering their native targets through resizing introduces a double-interpolation approximation.

The loss compares prediction and undegraded observation at each band's native sampling:

$$\mathcal{L} = \frac{1}{C} \sum_{c=1}^{C} \left\| R_c(\widehat{y}_c) - R_c(x_c^{\text{native}}) \right\|_1, \tag{6}$$

with R_c identity for finest-grid bands and bilinear resizing for coarser bands. Training band dropout retains at least one finest-resolution band and supports variable channel-set processing.

3. Experimental design

3.1. Training imagery and collection

The pretraining corpus combines native observations from Sentinel-2, Landsat-8/9, NAIP and VENuS (Table 1). Collection preserves each product's band definitions and radiometric convention; no cross-sensor high-resolution pair or harmonized cross-sensor training target is constructed.

Table 1: Pretraining imagery and collection sources. Band sampling is specified before Wald degradation; coarser bands are represented on each mission's finest working grid. NAIP is resampled to a canonical grid and retains scaled image digital numbers.

Mission	Bands	GSD (m)	Sources
Sentinel-2	12	10/20/60	CloudSEN12 / Planetary Computer
Landsat-8/9	7	30	USGS / Planetary Computer
NAIP	4	0.6	USDA / Planetary Computer
VENuS	12	5	Theia / EarthDaily

Sentinel-2 imagery combines a clear-image subset of CloudSEN12 Level-2A (Aybar et al., 2022) with additional Level-2A acquisitions queried at CloudSEN12 region-of-interest locations through the Microsoft Planetary Computer catalogue. Only spectral bands are retained; atmospheric auxiliary layers are excluded. The collection uses clear-image metadata and the scene classification layer to screen clouds and shadows. Raw image DN is converted using the fixed dataset scaling at loading.

Landsat-8/9 uses the Collection 2 Level-2 reflective bands from the Planetary Computer archive. Locations are sampled from CloudSEN12 centroids; the quality mask screens fill, cloud, dilated-cloud and shadow pixels. Raster scale and offset metadata are applied during retrieval, and the cache stores surface reflectance. NAIP supplies raw four-band aerial imagery from the USDA archive distributed by the same catalogue, covering the continental United States. The collection and tiling routines identify source acquisitions, select spatial windows, discard near-uniform patches and resample to the canonical cache grid. Its scaled digital numbers are not calibrated reflectance.

VENuS imagery is collected from the public Theia/EarthDaily Level-2A cloud-optimized GeoTIFF archive. The pipeline selects acquisitions at the mission's fixed monitoring sites using cloud-cover metadata, reads the spectral surface-reflectance bands and applies the product's scale and offset. Source scenes are tiled into the common cache format. Across missions, 512-pixel cache tiles retain the physical band descriptors while representing coarser bands on the finest working grid. Training and validation partitions are formed at patch level within each mission.

3.2. Multi-mission pretraining and scale-transfer

The mission sampler alternates mission-homogeneous batches while sharing every reconstruction module. Its recorded sampling weights are 0.2037, 0.1281, 0.6291 and 0.0391, in the order Sentinel-2, Landsat-8/9, NAIP and VENuS. NAIP contributes genuine fine-sampling observations to the shared prior; its detail is learned through its own reconstruction targets and reused through the common encoder and renderer. Scale-transfer therefore combines within-mission Wald supervision with cross-mission parameter sharing. Training uses four GPUs, one sample per GPU, bfloat16 mixed precision, gradient checkpointing and fused AdamW with peak learning rate 5×10^{-5}, warmup and cosine decay. BreizhSR and PROBA-V adaptation are separate supervised stages with held-out evaluation.

3.3. OpenSR cross-sensor evaluation

We score all five downloaded OpenSR suites: NAIP, SPOT, Spain crops and Spain urban at scale four, and VENuS at scale two. CSR receives Sentinel-2 RGBN input; evaluation scores RGB against the supplied harmonized reference. The official registration-aware OpenSR metrics characterize improvement (IM), hallucination (HA), omission and radiometric differences. Additional PSNR and SSIM are unaligned against that reference with range one. Cached author-released baseline outputs are rescored under the same harness where available. Training data and budgets are not matched across these methods.

Each method has one recorded operating point in the main comparison. OpenSR figures display the measured means without connecting them; green shading uses exact discrete point-estimate dominance. Scene-row bootstrap intervals use 10,000 resamples and seed zero to describe variation among the sampled scenes.

3.4. BreizhSR real-reference RGB reconstruction

BreizhSR provides paired Sentinel-2 and SPOT imagery (Okabayashi et al., 2024). We evaluate the released test metadata using the single closest-date Sentinel-2 acquisition, with ties resolved deterministically. Input RGB retains the fixed DN/10000 reflectance convention. SPOT display DN is histogram-matched to the input as reference preprocessing, aligning its display-scale distribution with the Sentinel-2 input. Fixed central crops have 74×74 input and 296×296 target pixels; edge examples smaller than these grids are excluded by input dimensions, with counts retained.

Paired uncertainty resamples 10-km Lambert-93 blocks. This physical-input protocol differs from the published percentile-normalized protocol, so the published learned-baseline numbers are not mixed with these measurements. The comparison supports RGB-only conclusions.

We extend this evaluation with supervised cross-sensor adaptation on the released training imagery. The published test partition is retained. Adaptation excludes its 10-km Lambert-93 blocks and shared parent footprints; validation blocks are drawn from the remaining training geography. Parent footprints spanning the training–validation boundary and exact input/reference file duplicates across partitions are excluded. A prespecified, spatially balanced subset of at most 256 validation scenes selects checkpoints. Fine-tuning and random initialization use the same optimizer, maximum update budget and early-stopping settings as the PROBA-V experiment. The loss extends its common-shift, brightness-corrected MSE to RGB, using one spatial translation for all channels and independent constant channel offsets. Input conditioning and the reference preprocessing are identical across all five tested methods.

3.5. External mission transfer: PROBA-V

PROBA-V supplies real 300-m and 100-m RED and NIR observations with quality masks (Märtens et al., 2019). It is absent from the declared CSR pretraining missions. We use the public labelled partition as an unadapted transfer evaluation; neither model training nor preprocessing is fitted to those references. Hidden challenge-test targets are unavailable and are excluded. For each scene, a single low-resolution image is selected by maximum clear-mask fraction, with lexical ties, before reading the high-resolution reference. Bicubic, Lanczos and CSR receive that same observation. This is a single-image evaluation, distinct from the original multi-image challenge.

All images retain the fixed benchmark encoding DN/65535. This preserves the benchmark digital-number scale. RED and NIR are conditioned with their instrument centre wavelengths and bandwidths; the MTF is prespecified at 0.2, with 0.1/0.3 sensitivity on a fixed initial subset. No reference-driven MTF selection is performed. The official clear-mask cPSNR searches 49 relative shifts after a three-pixel centre crop and removes a constant radiometric bias. We also report direct unaligned clear-mask PSNR and MAE to retain the alignment and radiometric discrepancies relevant to quantitative use. The input and reference may represent different acquisition times (Nguyen et al., 2021).

For adaptation, a prespecified seed partitions labelled scenes into approximately 70% training, 15% validation and 15% test. Scene identities and exact duplicate selected-input or reference hashes are grouped to prevent cross-partition duplication. We fine-tune CSR and train an identical architecture from random initialization using the same single-image inputs, training budget, optimizer and loss. A clear-mask, 49-shift, bias-corrected MSE handles registration and brightness differences during training. Full-image validation cPSNR selects checkpoints, with early stopping; both selections are frozen before opening test reference arrays for scoring. The original unadapted checkpoint and interpolation are evaluated on those identical test scenes. The split contains 812 training, 174 validation and 174 test scenes. Both trainable variants use AdamW at learning rate 0.00002, weight decay 0.0001, batch size 2 and 32-pixel input crops, with a maximum of 5000 updates. Validation is performed every 250 updates; 6 checks without at least 0.005 dB improvement stop training. This design evaluates supervised mission adaptation, separately from zero-shot transfer and the multi-image challenge.

4. Results

4.1. Cross-sensor fidelity and recovered structure

CSR has the largest recorded mean structural improvement on the four scale-four suites (Table 2). Its reduced omission indicates that it recovers structure left out by interpolation and conservative learned reconstructions.

Table 2: Matched OpenSR evaluation. IM: improvement; HA: hallucination; OM: omission. Pixel PSNR/SSIM are unaligned against the supplied harmonized reference; PSNR uses reflectance range one. Each row is an arithmetic mean over the same scenes.

Suite	n	s	Method	IM	HA	OM	PSNR	SSIM
NAIP	62	4	Bicubic	0.0421	0.0398	0.9182	38.64	0.9111
NAIP	62	4	CSR	0.2233	0.2937	0.4830	36.93	0.8917
SPOT	9	4	Bicubic	0.0402	0.0365	0.9233	34.48	0.8513
SPOT	9	4	CSR	0.2319	0.2693	0.4988	32.51	0.8146
VENuS	59	2	Bicubic	0.2866	0.1590	0.5544	39.79	0.9376
VENuS	59	2	CSR	0.2283	0.6052	0.1664	33.00	0.8161
Spain crops	28	4	Bicubic	0.0757	0.0632	0.8611	35.03	0.8544
Spain crops	28	4	CSR	0.2520	0.2300	0.5180	33.18	0.8235
Spain urban	20	4	Bicubic	0.0780	0.0621	0.8599	30.52	0.7298
Spain urban	20	4	CSR	0.2825	0.2383	0.4792	28.57	0.6944

Figure 2 locates this structural-recovery advantage in the omission–improvement plane. Both scores are interpreted together with hallucination because they decompose the same structure errors.

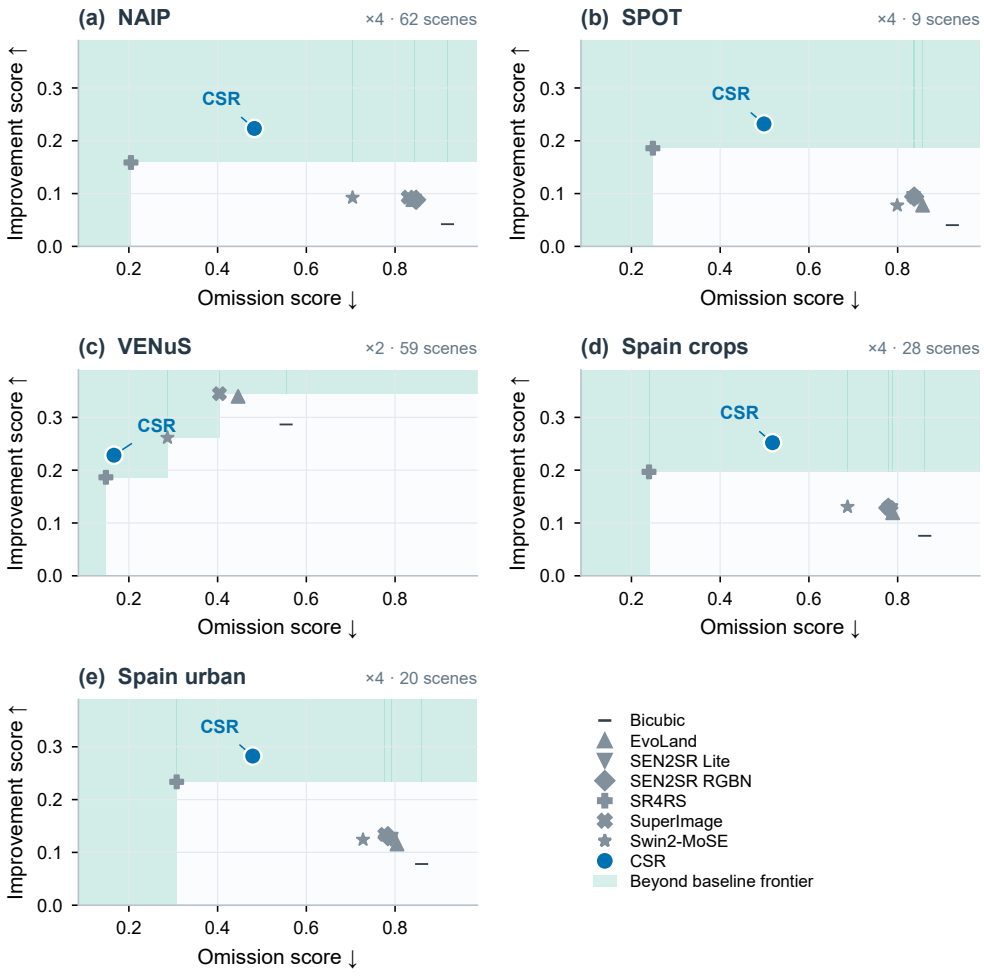


Figure 2: **CSR recovers structure omitted by conservative reconstructions.** Mean improvement versus omission on the same OpenSR scenes. Green marks extensions of the discrete baseline frontier. One trained CSR model; hallucination is reported separately.

Recovered structure also changes the pixel-fidelity balance. CSR lies on the observed mean PSNR–IM frontier for 4 of five suites (Figure 3), while several baselines retain higher PSNR. The scale-two VENuS suite has a different ordering.

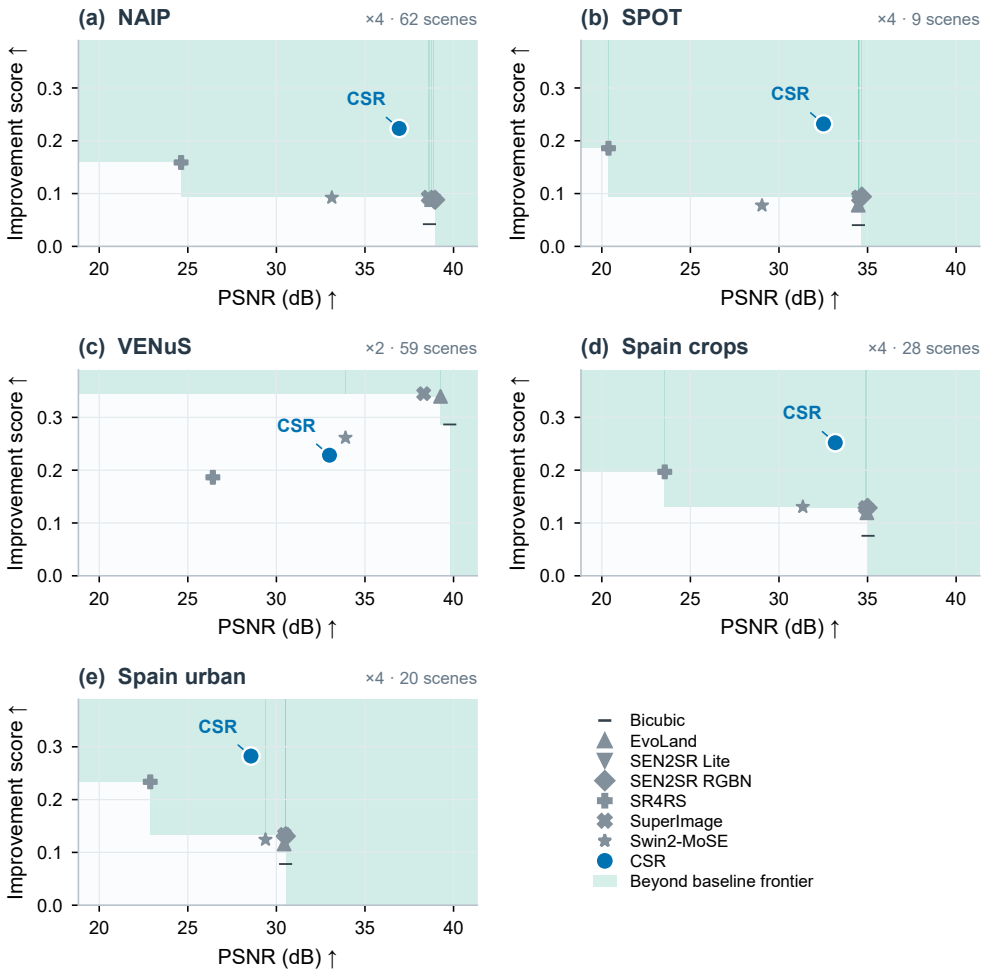


Figure 3: PSNR and structural improvement for CSR and available baselines. Green indicates extensions of the discrete baseline frontier. All markers represent measured method settings; no interpolated guide is drawn.

Figure 4 isolates improvement and its scene-level uncertainty. All four scale-four suites show the same ordering of CSR relative to the available methods; unavailable baseline settings remain explicit.

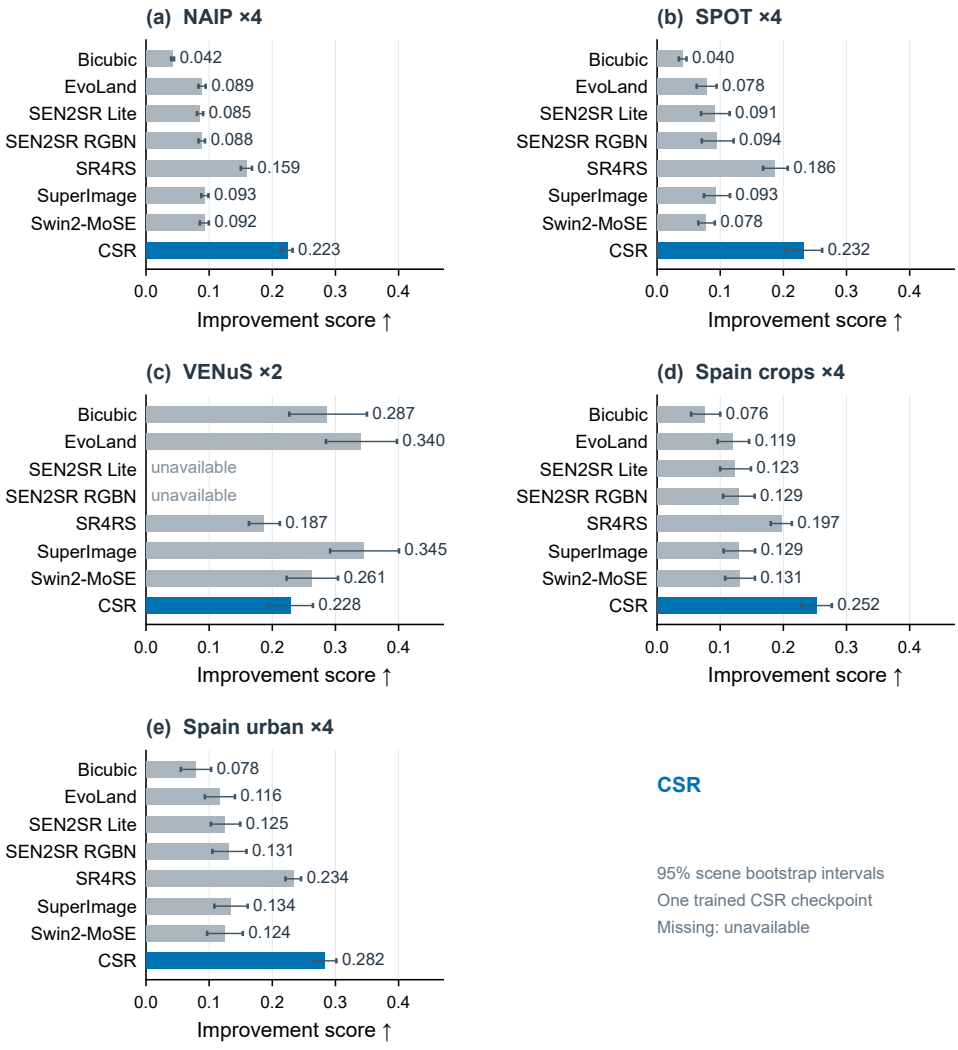


Figure 4: OpenSR improvement alone, with CSR highlighted and a shared metric axis. Whiskers are pointwise 95% scene-row bootstrap intervals for one trained model. Unavailable VENuS baseline settings are labelled.

The RGB comparisons in Figure 5 show how the methods differ in their reconstruction of spatial patterns. A common display stretch exposes the differences against the supplied harmonized reference.

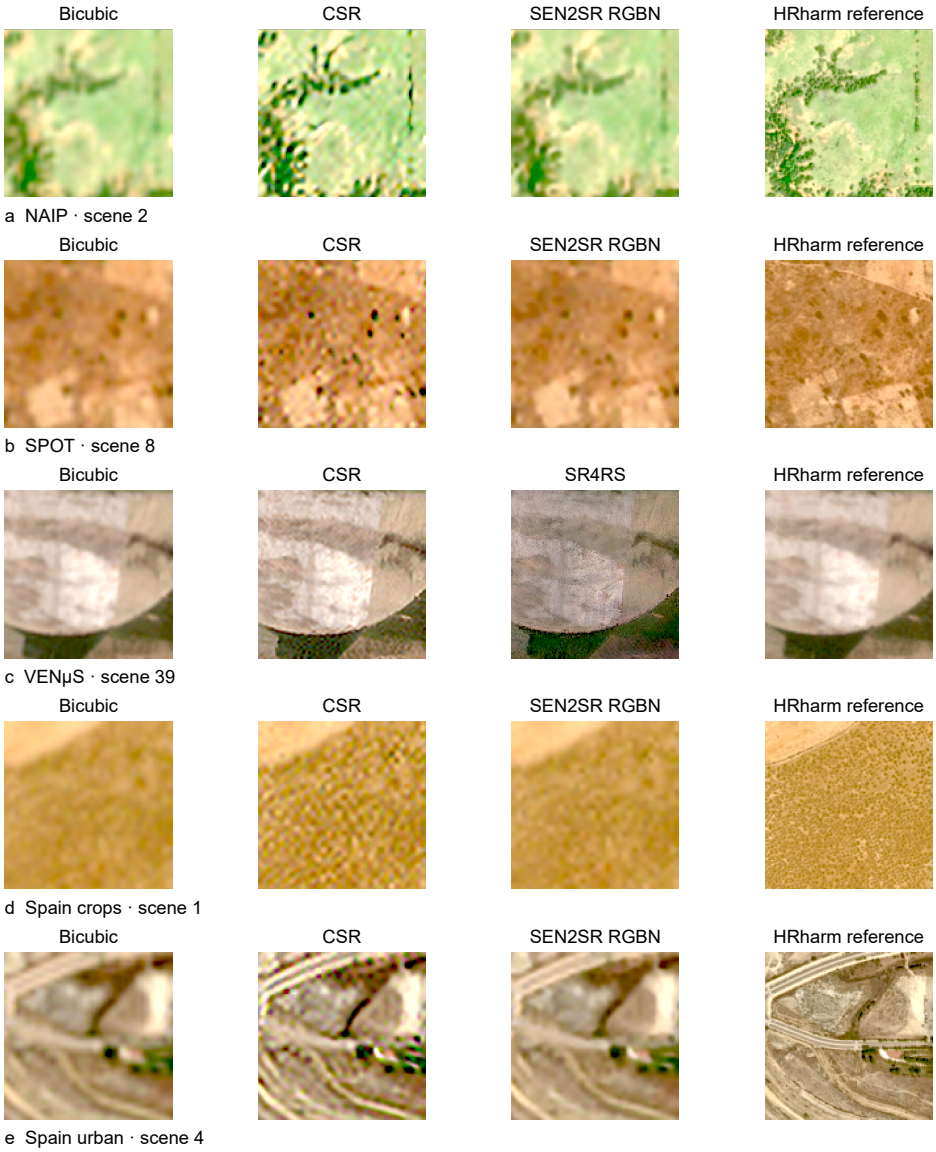


Figure 5: RGB reconstructions from bicubic, CSR, an available learned baseline and the harmonized reference. Shared central crops and reference-derived stretches. These previously selected illustrative scenes are not random samples; selection identifiers and the audit are retained.

4.2. BreizhSR real-reference reconstruction and adaptation

Table 3: Matched real-reference adaptation on the retained published BreizhSR test scenes. All methods use the same single-image inputs and histogram-matched reference. Checkpoints are selected using spatially separate validation; pretrained adaptation and random initialization share the maximum update budget. Corrected PSNR compensates a common RGB translation and per-band brightness shifts; per-metric coverage is reported in the appendix.

Method	n	IM	HA	Omission	PSNR	cPSNR
Bicubic	14823	0.0725	0.0784	0.8491	32.35	33.87
Lanczos	14823	0.0728	0.0789	0.8483	32.34	33.87
CSR pretrained	14823	0.1507	0.2583	0.5910	30.18	31.50
CSR adapted	14823	0.1961	0.1548	0.6491	32.68	34.33
Random init.	14823	0.0788	0.0833	0.8379	32.43	33.92

Table 3 compares pretrained CSR, adapted CSR, matched random initialization and interpolation on identical real-reference pairs. Of 14831 released test pairs, 6 have insufficient crop dimensions and 2 have wholly zero processed references, leaving 14823 evaluated scenes. Undefined metrics remain unavailable with their finite-value counts retained. The spatially separate adaptation partitions contain 10083 training and 2645 validation examples; 256 prespecified validation examples select checkpoints before test scoring.

Adaptation increases corrected PSNR by 2.82 dB over the pretrained checkpoint, 0.46 dB over bicubic (95% spatial-block interval [0.40, 0.52]) and 0.41 dB over random initialization (interval [0.36, 0.46]). Direct unaligned PSNR also improves: the gains are 0.33 dB over bicubic and 0.25 dB over random initialization. The pretrained model therefore supplies a useful starting point for learning from paired Sentinel-2 and SPOT observations, improving both corrected and direct fidelity beyond fitting the architecture from scratch.

Adapted CSR also achieves the largest mean structural improvement in this matched comparison. IM rises from 0.1507 to 0.1961, while HA falls from 0.2583 to 0.1548. The IM gain is 0.0454 (interval [0.0331, 0.0582]). Adaptation concentrates the reconstruction on reference-supported structure, with omission increasing from 0.5910 to 0.6491 while remaining below bicubic's 0.8491. Hallucination remains higher than interpolation and random initialization. These scores distinguish the detail-rich pretrained

reconstruction from the adapted model's stronger fidelity and structural improvement. Complete metrics, checkpoint selections and paired intervals appear in Tables A.11–A.13. The protocol retains acquisition-time differences and reference-only radiometric matching; its absolute scores describe this physical-input comparison.

Figure 6 shows fixed held-out examples under a shared display procedure.

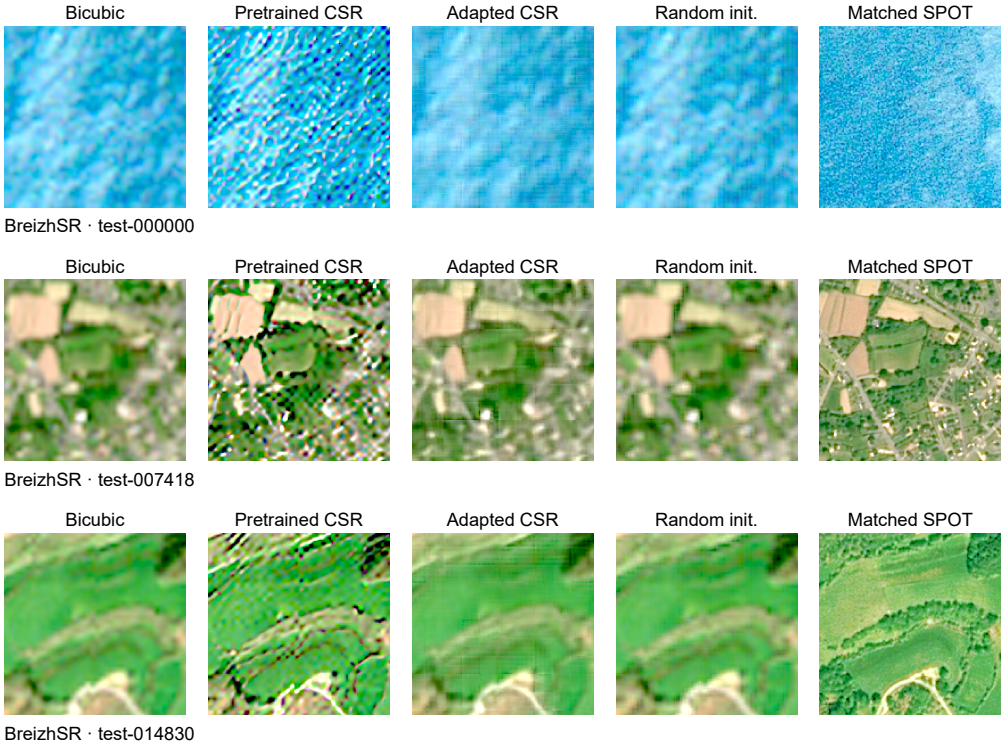


Figure 6: **Task adaptation changes the reconstruction of real Sentinel-2–SPOT pairs.** Pretrained, adapted and randomly initialized CSR share the same architecture; bicubic and the matched SPOT reference provide context. Fixed first, middle and last eligible metadata examples use shared input-derived stretches and central crops. One adaptation seed; examples are selected independently of performance.

4.3. Multi-mission reconstruction scope

Figure 7 illustrates native-to-finer inference with the same checkpoint across Sentinel-2, Landsat-8/9, NAIP and VENuS. It includes infrared and

shortwave-infrared composites and a noninteger output ratio. CSR accepts these different band arrangements and physical descriptors through its shared encoder and renderer.

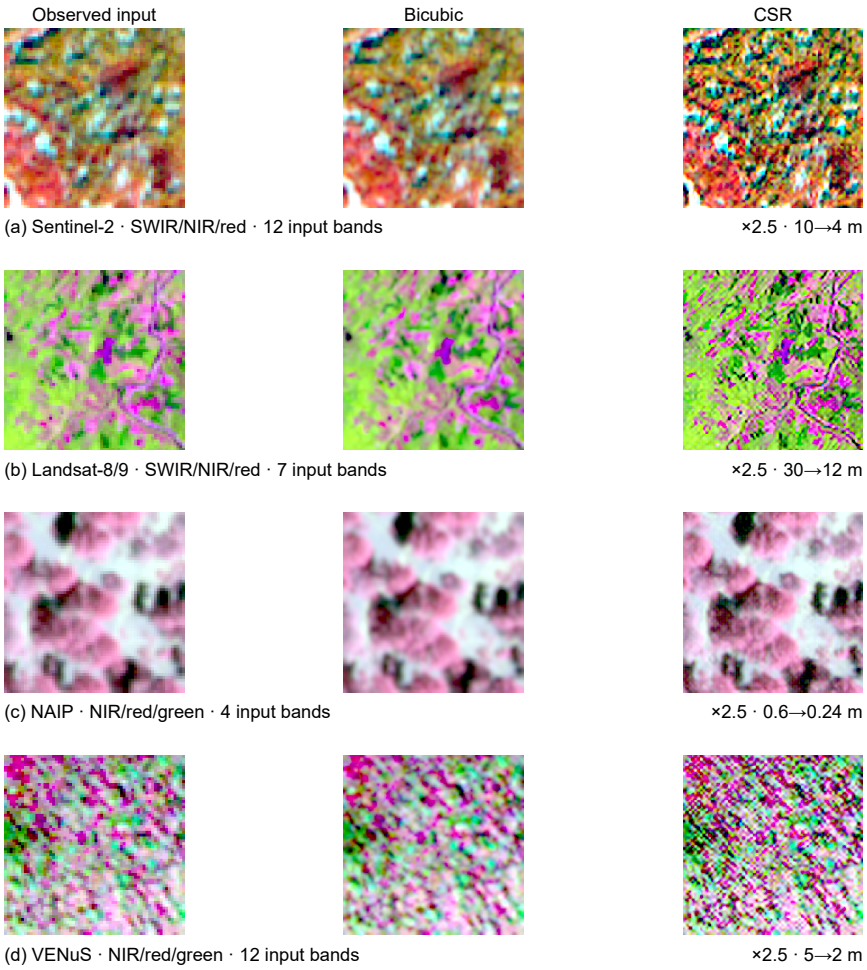


Figure 7: **One parameterized architecture processes heterogeneous optical missions.** Fixed native patches on the mission working grid are reconstructed at a noninteger grid ratio with bicubic and CSR. Bandwise physical ratios retain their native sampling differences. Observed inputs and all predictions share input-derived bandwise stretches; composites expose infrared and shortwave-infrared channels. These native-to-finer illustrations have no fine-resolution reference and demonstrate inference scope rather than measured recovery accuracy.

4.4. External-mission adaptation

Table 4: PROBA-V single-input reconstruction from 300 m to 100 m. The upper panel scores every public labelled scene without adaptation; the lower panel compares all methods on the same held-out test scenes after validation-only model selection. cPSNR corrects registration and bias; PSNR, MAE and RMSE use the direct clear reference mask. The cohorts have different coverage and must not be compared as matched samples.

Band	n	Method	cPSNR	PSNR	MAE	RMSE
Public labelled partition: unadapted transfer						
RED	594	Bicubic	47.48	41.51	0.0080	0.0095
RED	594	Lanczos	47.56	41.51	0.0080	0.0095
RED	594	CSR pretrained	43.97	39.09	0.0096	0.0121
NIR	566	Bicubic	45.21	39.54	0.0099	0.0118
NIR	566	Lanczos	45.28	39.54	0.0099	0.0118
NIR	566	CSR pretrained	41.66	37.03	0.0120	0.0153
Held-out test split: paired adaptation comparison						
RED	89	Bicubic	47.51	41.13	0.0087	0.0102
RED	89	Lanczos	47.60	41.13	0.0087	0.0102
RED	89	CSR pretrained	43.97	38.78	0.0101	0.0127
RED	89	CSR adapted	48.09	40.87	0.0088	0.0105
RED	89	Random initialization	47.64	41.05	0.0088	0.0103
NIR	85	Bicubic	45.40	39.80	0.0096	0.0115
NIR	85	Lanczos	45.47	39.80	0.0096	0.0115
NIR	85	CSR pretrained	42.11	37.39	0.0116	0.0149
NIR	85	CSR adapted	45.87	39.52	0.0097	0.0118
NIR	85	Random initialization	45.50	39.71	0.0096	0.0116

PROBA-V demonstrates adaptation of the same reconstruction architecture to a mission outside the pretraining pool. The lower panel of Table 4 compares the adapted model, its pretrained initialization, interpolation and random initialization on identical held-out scenes. The upper panel records the broader unadapted starting point across 1160 labelled scenes, with 0 exclusions. CSR's unadapted corrected PSNR is lower than bicubic by 3.51 dB in RED and 3.55 dB in NIR, providing a demanding starting point for mission adaptation. Figure 8 illustrates that initial reconstruction. Corrected cPSNR evaluates spatial fidelity after compensating for small registration and constant brightness differences; direct masked metrics retain those differences.

Together they distinguish learned reconstruction quality from acquisition-level alignment and radiometry.

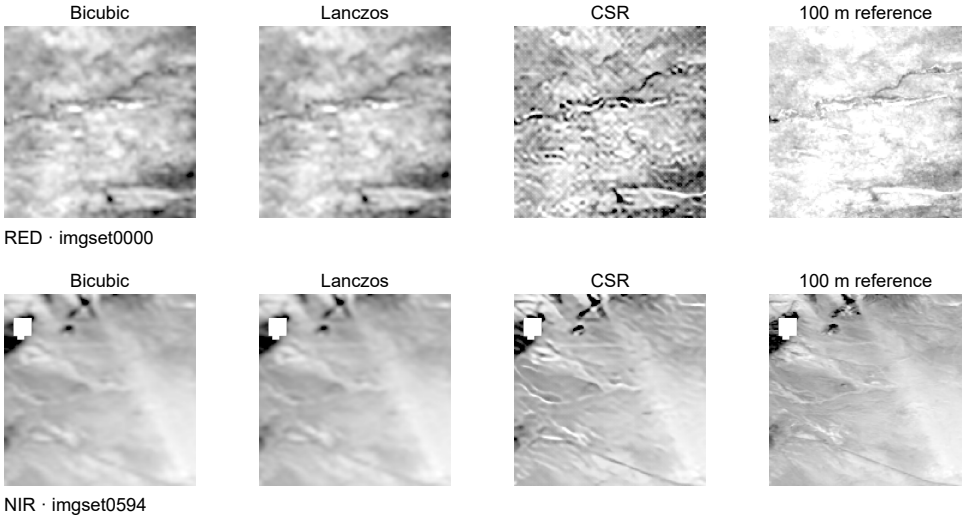


Figure 8: Prespecified first PROBA-V example for each band. Bicubic, Lanczos, CSR and the real 100-m reference share a fixed central crop and input-derived stretch. Invalid reference-mask pixels are blank. These unaligned illustrations retain temporal and registration differences.

The lower panel of Table 4 uses the held-out adaptation split. Fine-tuning increases cPSNR over the unadapted initialization by 4.12 dB in RED and 3.75 dB in NIR. It exceeds bicubic by 0.58 and 0.47 dB, and the random-initialization control by 0.45 and 0.37 dB, respectively (Table 4). Paired scene/duplicate-group bootstrap intervals are reported in Table A.10. The validation-selected fine-tuned checkpoint occurs at step 1750, while the control selects step 3250 (Table A.9). The consistent advantage over random initialization demonstrates the practical value of the pretrained scale-transfer prior for this external optical mission.

Adaptation improves registration- and bias-corrected spatial fidelity, the benchmark's primary criterion. Direct PSNR relative to bicubic changes by -0.26 dB in RED and -0.28 dB in NIR; the direct metrics therefore retain a small interpolation advantage. Figure 9 contrasts the zero-shot and adapted reconstructions on fixed held-out examples.

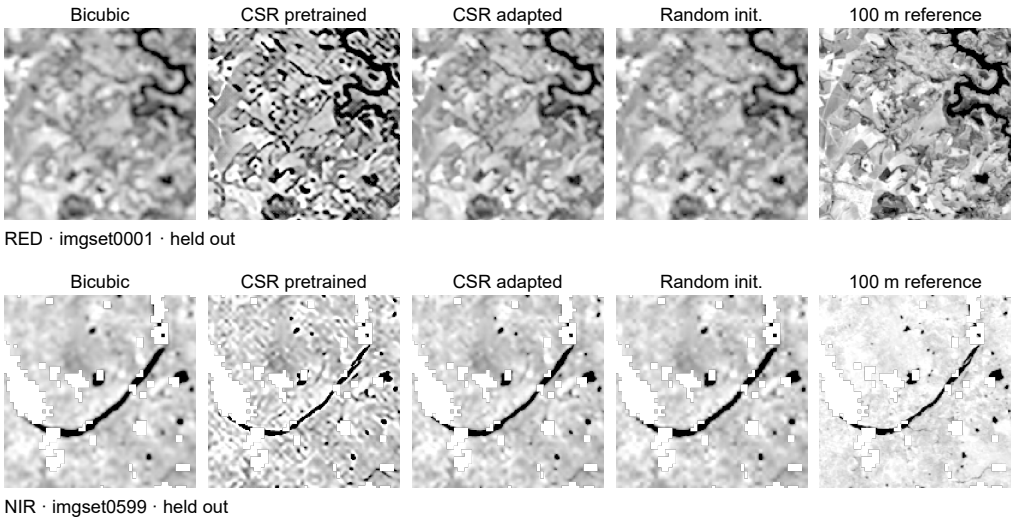


Figure 9: **Supervised adaptation changes reconstruction on an external mission.** The first held-out test example in each band is shown for bicubic, pretrained CSR, adapted CSR, the random-initialization control and the real 100-m reference. Shared input-derived stretches, fixed central crops and blank invalid reference pixels; no image is selected by model performance.

4.5. Hallucination and spatial error patterns

The structural scores identify CSR's principal strength: recovering fine patterns that other reconstructions omit. Figure 10 complements the omission view by showing the accompanying hallucination scores. CSR's recovered-detail advantage is strongest in the scale-four suites; the joint IM–HA view distinguishes that advantage from the more conservative reconstruction settings. In remote sensing, this distinction matters because field boundaries and small built structures can become more visible while individual reference pixels retain radiometric or positional differences.

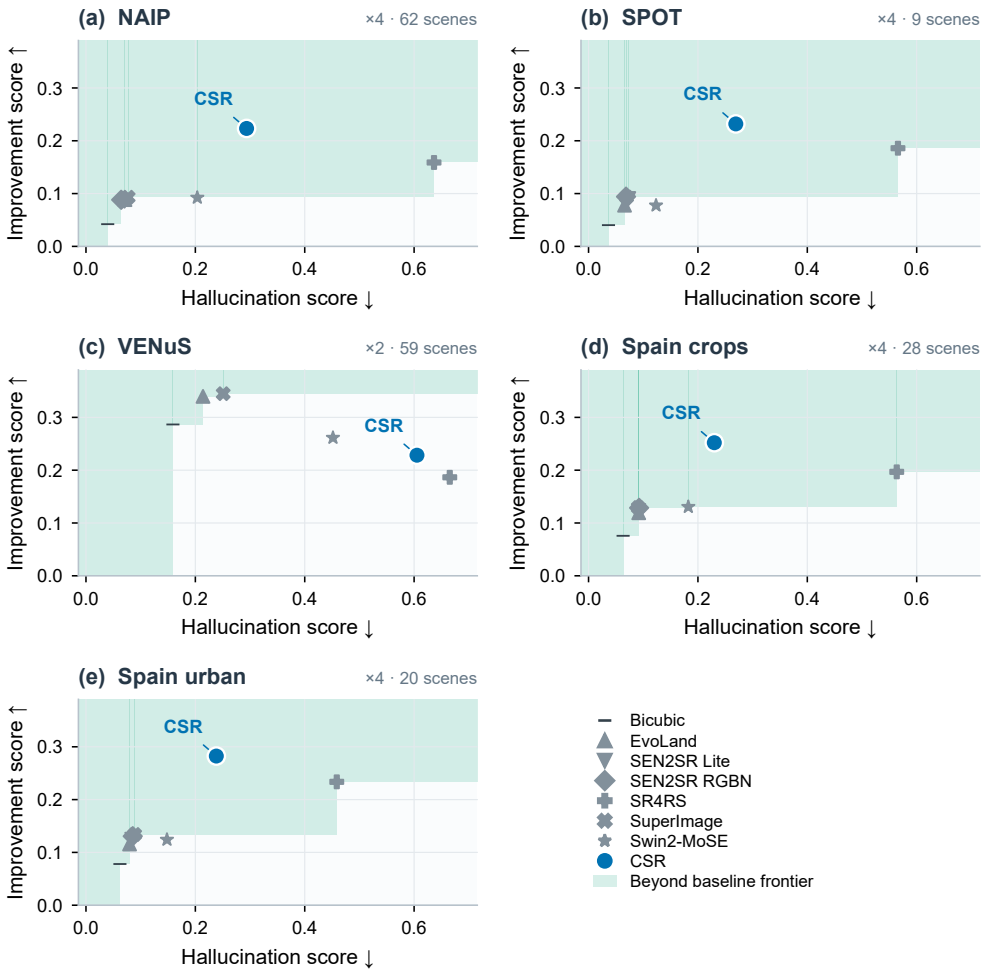


Figure 10: Improvement and hallucination for the measured OpenSR method settings. Green marks extensions of the discrete baseline frontier. The two axes separate recovered structure from introduced structure.

Figure 11 localizes pixel discrepancies on the same fixed scenes used in the qualitative comparisons. All error panels show mean absolute RGB differences from the harmonized reference with a common color range. In the NAIP crop, error hotspots follow vegetation boundaries and narrow dark features. The SPOT crop combines a broad local offset with mismatched fine texture. In the urban crop, bright linear infrastructure and intersecting boundaries dominate the visible hotspots in both reconstructions. The

strongest visible hotspots therefore occur at high-contrast transitions and in regions with local radiometric disagreement. The reference-gradient stratification in Table 5 quantifies this concentration for each illustrated crop using the same edge mask across methods. These unaligned maps preserve the effects of registration, acquisition timing and reference harmonization alongside reconstruction error, whereas the OpenSR structure metrics evaluate registered patterns. Reading them together explains why recovered structure and pixel PSNR can favor different methods.

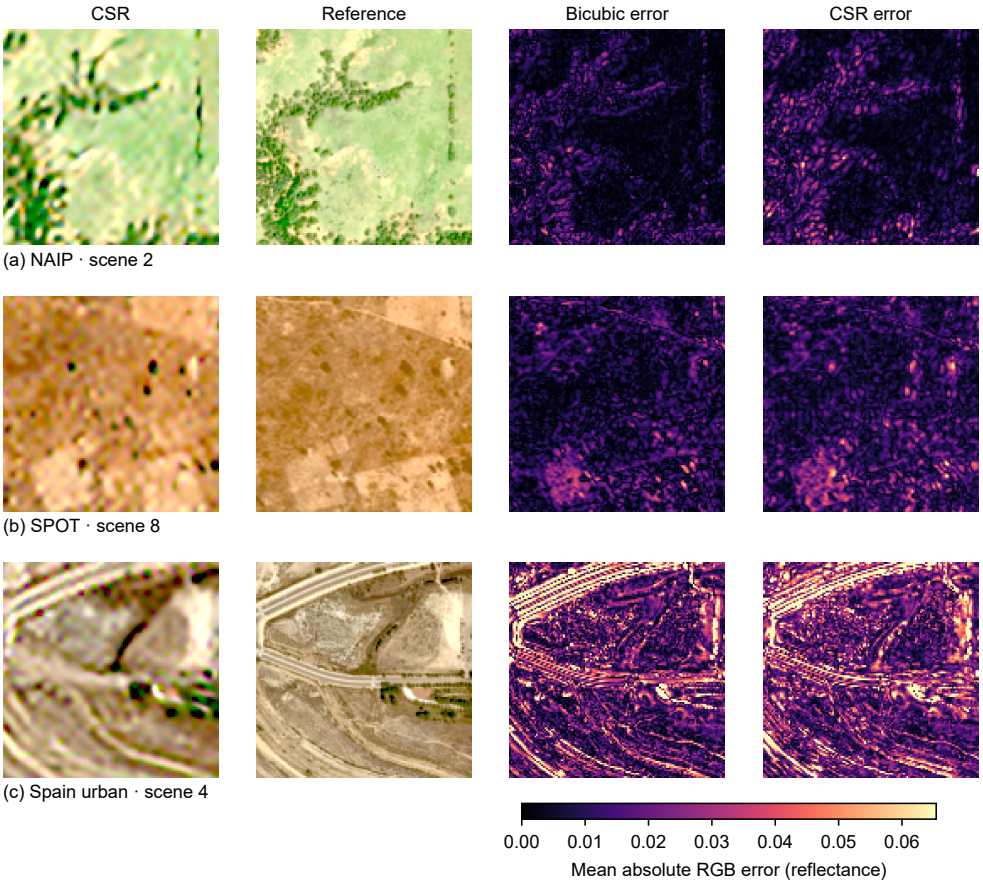


Figure 11: **Spatial error is concentrated around fine boundaries and locally different surface patterns.** CSR and the harmonized reference accompany bicubic and CSR absolute-error maps on the same fixed central crops. A shared color scale reports mean absolute RGB error in range-one reflectance; its upper limit is the pooled 98th percentile, with larger errors saturated. Source scene identifiers and crop definitions match the qualitative panel.

Table 5: Spatial error stratification for the fixed crops in Figure 11. Edges are the highest quartile of reference gradient magnitude, shared across methods. MAE is mean absolute RGB error in range-one reflectance; edge share is the fraction of summed absolute error in that quarter of the crop.

Crop	Method	Edge MAE	Other MAE	Edge share
naip	Bicubic	0.0093	0.0049	38.6%
naip	CSR	0.0126	0.0063	39.9%
spot	Bicubic	0.0109	0.0083	30.5%
spot	CSR	0.0134	0.0104	30.1%
spain urban	Bicubic	0.0337	0.0215	34.4%
spain urban	CSR	0.0381	0.0240	34.6%

Frequency composition provides a complementary form of control: it reduces low-frequency and degradation-back discrepancies relative to the raw decoder residual. Its smooth MTF mask preserves an approximate observation anchor while allowing finer spatial detail to enter. The same-weight coordinate and frequency sensitivities in Table C.20 explain this inference behavior in detail.

5. Discussion

CSR provides a reusable reconstruction interface across optical missions, band arrangements and sampling grids. Physical descriptors let a shared feature extractor distinguish bands, spatial–spectral attention exchanges information across the supplied channel set, and coordinate rendering decouples the requested grid from a fixed output head. Native supervision makes training feasible without paired cross-sensor high-resolution targets. The multi-mission illustrations demonstrate this operational scope; real-reference evaluation quantifies the reconstruction choices it produces.

5.1. Recovered structure and pixel fidelity

CSR's highest mean improvement on the OpenSR scale-four suites demonstrates its strength in recovering finer spatial structure. The real-reference BreizhSR evaluation extends this evidence to paired Sentinel-2 and SPOT imagery. The omission view makes the structural advantage explicit. Improvement, hallucination and pixel fidelity together characterize the resulting reconstruction, since a sharper image need not agree more closely with every reference pixel.

5.2. Transfer and adaptation

Continuous band descriptors support reuse without inserting a mission-specific output head. They also expose how a new instrument differs in bandpass, sampling and spatial response. BreizhSR tests supervised adaptation within the Sentinel-2 input interface, while PROBA-V extends it to a mission absent from pretraining. Both use real references: unadapted reconstruction supplies the starting point, and validation-selected fine-tuning tests reuse of the pretrained model. An equal-budget random-initialization control separates adaptation from simply fitting a task-specific network. The higher held-out corrected fidelity of pretrained adaptation on both benchmarks supports the reuse of the scale-transfer prior and a foundation-model role within geospatial super-resolution: a shared multi-mission backbone can be adapted to a new optical mission without redesigning its input or output heads. The resulting backbone supports mission adaptation through continued learning with a common band-conditioned interface.

5.3. Observation consistency and evaluation context

The benchmarks capture complementary remote-sensing conditions. OpenSR and BreizhSR assess detail against finer cross-sensor references, while PROBA-V evaluates adaptation using real coarse and fine observations. Reference harmonization, acquisition timing and registration each influence pixel differences, making spatial structure and radiometric fidelity complementary rather than interchangeable measures.

CSR's frequency composition anchors low-frequency content to the observation with a smooth MTF response. Future work can strengthen this physical consistency, incorporate more complete spectral response functions and evaluate geographically separated acquisition inventories. The present experiments use one pretrained checkpoint and one adaptation seed; matched mission-removal training controls and further adaptation seeds would quantify the contribution of each training source and robustness across optimization runs.

6. Conclusion

CSR combines physical band conditioning, shared spatial–spectral processing and coordinate rendering to super-resolve heterogeneous optical imagery with one parameterized architecture. Scale-transfer combines multi-mission native supervision with a shared physically conditioned detail prior,

supplying a practical pretraining route without cross-sensor high-resolution pairs. CSR delivers the strongest measured OpenSR improvement across the scale-four suites, with increased structural improvement and reduced omission on real-reference BreizhSR pairs. Real-reference adaptation improves corrected fidelity over interpolation and equal-budget random initialization on both BreizhSR and PROBA-V; BreizhSR also improves direct pixel fidelity. This demonstrates reuse of the pretrained model for paired cross-sensor reconstruction and for an external mission. Together with the mission illustrations, this connects multi-mission pretraining to a transferable geospatial super-resolution backbone. The evaluation establishes both the model's flexible scope and the task-dependent balance between fidelity and recovered detail.

7. Data and code availability

Recorded per-scene and per-band measurements, checkpoint and input hashes, coverage/exclusions, environment snapshots and source run identifiers accompany the evaluation data. Programmatic scripts regenerate numerical result macros, tables and bibliography entries; figure scripts retain sampling and display audits. The public datasets include BreizhSR (`https://zenodo.org/records/11551220`), PROBA-V (`https://zenodo.org/records/6327426`) and the LS2S2 preview (`https://doi.org/10.5281/zenodo.15471890`). Public code and model release details will be supplied before submission.

Appendix A. Complete cross-sensor and native diagnostics

Table A.6: All cached methods rescored under the same OpenSR harness. Baselines differ in training data, architecture and protocol; these are checkpoint comparisons, not controlled architecture ablations. Missing values remain unavailable.

Suite/method	n	IM	HA	OM	Refl.	Spectral	Spatial	Synth.
NAIP/Bicubic	62	0.0421	0.0398	0.9182	0.0016	0.4320	0.0006	0.0013
NAIP/EvoLand	62	0.0886	0.0713	0.8401	0.0017	0.7379	0.0024	0.0023
NAIP/SEN2SR Lite	62	0.0854	0.0705	0.8441	0.0013	0.4231	0.0000	0.0023
NAIP/SEN2SR RGBN	62	0.0882	0.0644	0.8474	0.0012	0.4366	0.0000	0.0022
NAIP/SR4RS	62	0.1589	0.6364	0.2047	0.0346	3.0728	1.0136	0.0137
NAIP/SuperImage	62	0.0931	0.0772	0.8297	0.0025	1.5309	0.0000	0.0026
NAIP/Swin2-MoSE	62	0.0923	0.2035	0.7042	0.0181	2.0056	0.2100	0.0036
NAIP/CSR	62	0.2233	0.2937	0.4830	0.0021	0.6483	0.0000	0.0062
SPOT/Bicubic	9	0.0402	0.0365	0.9233	0.0033	0.5423	0.0000	0.0027
SPOT/EvoLand	9	0.0780	0.0658	0.8562	0.0030	0.7481	0.0022	0.0045
SPOT/SEN2SR Lite	9	0.0907	0.0735	0.8358	0.0026	0.4981	0.0000	0.0048
SPOT/SEN2SR RGBN	9	0.0942	0.0685	0.8373	0.0022	0.5112	0.0000	0.0048
SPOT/SR4RS	9	0.1859	0.5658	0.2483	0.0557	3.8256	1.0305	0.0198
SPOT/SuperImage	9	0.0934	0.0696	0.8370	0.0032	0.9342	0.0000	0.0050
SPOT/Swin2-MoSE	9	0.0775	0.1234	0.7991	0.0242	1.5968	0.2017	0.0053
SPOT/CSR	9	0.2319	0.2693	0.4988	0.0043	0.8466	0.0054	0.0117
VENuS/Bicubic	59	0.2866	0.1590	0.5544	0.0024	0.8134	0.0000	0.0045
VENuS/EvoLand	59	0.3398	0.2140	0.4462	0.0021	1.1407	0.0010	0.0052
VENuS/SR4RS	59	0.1865	0.6652	0.1483	0.0188	4.1841	0.7615	0.0121
VENuS/SuperImage	59	0.3449	0.2509	0.4042	0.0035	2.2000	0.1206	0.0057
VENuS/Swin2-MoSE	59	0.2614	0.4516	0.2869	0.0152	3.6707	0.1594	0.0075
VENuS/CSR	59	0.2283	0.6052	0.1664	0.0029	1.1982	0.0023	0.0103
Spain crops/Bicubic	28	0.0757	0.0632	0.8611	0.0019	0.4987	0.0000	0.0027
Spain crops/EvoLand	28	0.1194	0.0917	0.7889	0.0019	0.8328	0.0049	0.0040
Spain crops/SEN2SR Lite	28	0.1235	0.0919	0.7846	0.0016	0.4978	0.0000	0.0039
Spain crops/SEN2SR RGBN	28	0.1287	0.0924	0.7789	0.0013	0.5040	0.0000	0.0040
Spain crops/SR4RS	28	0.1969	0.5635	0.2396	0.0447	3.6331	1.0144	0.0193
Spain crops/SuperImage	28	0.1293	0.0909	0.7798	0.0026	1.6292	0.0000	0.0041
Spain crops/Swin2-MoSE	28	0.1306	0.1824	0.6870	0.0211	1.9270	0.1851	0.0057
Spain crops/CSR	28	0.2520	0.2300	0.5180	0.0023	0.7388	0.0000	0.0084
Spain urban/Bicubic	20	0.0780	0.0621	0.8599	0.0036	0.7984	0.0000	0.0051
Spain urban/EvoLand	20	0.1159	0.0797	0.8044	0.0030	1.1346	0.0010	0.0068
Spain urban/SEN2SR Lite	20	0.1250	0.0837	0.7913	0.0029	0.7782	0.0000	0.0070
Spain urban/SEN2SR RGBN	20	0.1307	0.0855	0.7838	0.0024	0.7824	0.0000	0.0072
Spain urban/SR4RS	20	0.2336	0.4586	0.3078	0.0396	3.2631	0.9743	0.0247
Spain urban/SuperImage	20	0.1340	0.0891	0.7770	0.0033	1.6787	0.0000	0.0075
Spain urban/Swin2-MoSE	20	0.1241	0.1481	0.7278	0.0155	2.6353	0.2203	0.0090
Spain urban/CSR	20	0.2825	0.2383	0.4792	0.0045	1.2346	0.0020	0.0153

Table A.7: All cached methods rescored under the same OpenSR harness. Baselines differ in training data, architecture and protocol; these are checkpoint comparisons, not controlled architecture ablations. Missing values remain unavailable.

Suite/method	n	PSNR	SSIM	MAE	RMSE	SAM	ERGAS	Zero frac.
NAIP/Bicubic	62	38.6394	0.9111	0.0087	0.0126	1.4486	3.0268	0.0000
NAIP/EvoLand	62	38.7523	0.9139	0.0086	0.0125	1.3139	2.9857	0.0000
NAIP/SEN2SR Lite	62	38.8699	0.9158	0.0086	0.0123	1.5335	2.9511	0.0000
NAIP/SEN2SR RGBN	62	38.9587	0.9178	0.0085	0.0122	1.4346	2.9144	0.0000
NAIP/SR4RS	62	24.6199	0.6104	0.0458	0.0637	3.3610	14.3958	0.0000
NAIP/SuperImage	62	38.5719	0.9145	0.0090	0.0126	2.2388	3.0902	0.0000
NAIP/Swin2-MoSE	62	33.1299	0.8755	0.0195	0.0232	2.4088	5.7082	0.0000
NAIP/CSR	62	36.9338	0.8917	0.0104	0.0153	2.9650	3.8115	0.0000
SPOT/Bicubic	9	34.4834	0.8513	0.0155	0.0224	1.2898	3.9702	0.0000
SPOT/EvoLand	9	34.4732	0.8545	0.0154	0.0224	1.1955	3.9686	0.0000
SPOT/SEN2SR Lite	9	34.5467	0.8600	0.0152	0.0220	1.4721	3.9029	0.0000
SPOT/SEN2SR RGBN	9	34.6781	0.8642	0.0150	0.0218	1.3399	3.8604	0.0000
SPOT/SR4RS	9	20.3770	0.4872	0.0729	0.0986	4.0578	15.4621	0.0000
SPOT/SuperImage	9	34.4864	0.8599	0.0154	0.0222	1.8561	3.9523	0.0000
SPOT/Swin2-MoSE	9	29.0421	0.8242	0.0294	0.0365	1.7508	5.9168	0.0000
SPOT/CSR	9	32.5135	0.8146	0.0198	0.0294	3.7068	5.3474	0.0000
VENuS/Bicubic	59	39.7862	0.9376	0.0081	0.0119	2.2360	7.7184	0.0000
VENuS/EvoLand	59	39.2646	0.9319	0.0084	0.0126	2.1052	8.1995	0.0000
VENuS/SR4RS	59	26.4139	0.6398	0.0356	0.0509	4.6742	30.6657	0.0000
VENuS/SuperImage	59	38.3138	0.9231	0.0092	0.0136	3.4023	9.0735	0.0000
VENuS/Swin2-MoSE	59	33.9010	0.8821	0.0172	0.0215	4.1835	14.4188	0.0000
VENuS/CSR	59	32.9971	0.8161	0.0150	0.0246	5.8256	16.6568	0.0000
Spain crops/Bicubic	28	35.0273	0.8544	0.0129	0.0191	1.9350	4.4336	0.0000
Spain crops/EvoLand	28	34.9629	0.8555	0.0129	0.0192	1.7844	4.4846	0.0000
Spain crops/SEN2SR Lite	28	34.9475	0.8548	0.0129	0.0192	2.0372	4.5339	0.0000
Spain crops/SEN2SR RGBN	28	35.0020	0.8576	0.0128	0.0191	1.9249	4.4895	0.0000
Spain crops/SR4RS	28	23.5538	0.5756	0.0574	0.0779	4.0430	15.0042	0.0000
Spain crops/SuperImage	28	34.8824	0.8546	0.0131	0.0193	2.7080	4.5932	0.0000
Spain crops/Swin2-MoSE	28	31.3474	0.8299	0.0238	0.0292	2.5045	6.3790	0.0000
Spain crops/CSR	28	33.1788	0.8235	0.0150	0.0231	3.6134	5.9141	0.0000
Spain urban/Bicubic	20	30.5216	0.7298	0.0221	0.0322	2.9483	7.0524	0.0000
Spain urban/EvoLand	20	30.4388	0.7330	0.0222	0.0325	2.8184	7.1450	0.0000
Spain urban/SEN2SR Lite	20	30.5149	0.7352	0.0220	0.0322	3.0799	7.1279	0.0000
Spain urban/SEN2SR RGBN	20	30.5300	0.7389	0.0219	0.0322	2.9428	7.0990	0.0000
Spain urban/SR4RS	20	22.8791	0.4553	0.0574	0.0799	3.9991	15.9315	0.0000
Spain urban/SuperImage	20	30.4609	0.7344	0.0222	0.0324	3.6862	7.1741	0.0000
Spain urban/Swin2-MoSE	20	29.3839	0.7024	0.0264	0.0356	3.5862	7.9950	0.0000
Spain urban/CSR	20	28.5655	0.6944	0.0263	0.0398	5.7848	9.1808	0.0000

Table A.8: All native-resolution diagnostic cells, excluding wholly zero targets. Filename-overlap removal is not a geographic independence audit. SAM excludes zero target spectra in the evaluation code. These patches are not a new independent benchmark.

Mission	s	Method	n	PSNR	MAE	SAM
Landsat-8/9	1.0	Bicubic	64	47.4529	0.0039	0.8821

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Table A.8: continued

Mission	s	Method	n	PSNR	MAE	SAM
Landsat-8/9	1.0	CSR	64	61.0949	0.0008	0.2189
Landsat-8/9	1.5	Bicubic	64	40.8930	0.0077	1.6732
Landsat-8/9	1.5	CSR	64	47.0649	0.0037	0.9355
Landsat-8/9	2.0	Bicubic	64	38.9984	0.0094	1.9951
Landsat-8/9	2.0	CSR	64	43.6570	0.0054	1.2646
Landsat-8/9	2.5	Bicubic	64	37.8580	0.0106	2.2112
Landsat-8/9	2.5	CSR	64	41.5960	0.0068	1.5065
Landsat-8/9	3.0	Bicubic	64	36.7522	0.0116	2.3996
Landsat-8/9	3.0	CSR	64	39.3714	0.0084	1.9705
Landsat-8/9	4.0	Bicubic	64	35.2785	0.0137	2.7822
Landsat-8/9	4.0	CSR	64	37.8427	0.0100	2.0812
NAIP	1.0	Bicubic	64	41.0077	0.0074	0.6580
NAIP	1.0	CSR	64	54.6464	0.0015	0.1597
NAIP	1.5	Bicubic	64	34.7326	0.0150	1.2687
NAIP	1.5	CSR	64	40.7152	0.0074	0.7349
NAIP	2.0	Bicubic	64	32.6661	0.0190	1.5502
NAIP	2.0	CSR	64	37.3537	0.0109	1.0074
NAIP	2.5	Bicubic	64	31.4549	0.0217	1.7411
NAIP	2.5	CSR	64	35.2573	0.0137	1.2107
NAIP	3.0	Bicubic	64	30.6095	0.0239	1.8925
NAIP	3.0	CSR	64	33.6738	0.0164	1.3939
NAIP	4.0	Bicubic	64	29.2577	0.0281	2.1740
NAIP	4.0	CSR	64	31.8168	0.0203	1.6521
NAIP	6.0	Bicubic	64	27.7213	0.0335	2.5277
NAIP	6.0	CSR	64	29.6853	0.0259	2.0274
NAIP	8.0	Bicubic	64	26.7456	0.0372	2.7694
NAIP	8.0	CSR	64	28.4552	0.0297	2.2832
Sentinel-2	1.0	Bicubic	64	41.3544	0.0073	1.4962
Sentinel-2	1.0	CSR	64	44.0364	0.0048	1.1254
Sentinel-2	1.5	Bicubic	64	38.5694	0.0103	1.9897
Sentinel-2	1.5	CSR	64	42.8557	0.0058	1.3038
Sentinel-2	2.0	Bicubic	64	36.5604	0.0131	2.3948
Sentinel-2	2.0	CSR	64	41.7807	0.0066	1.4587
Sentinel-2	2.5	Bicubic	64	35.2598	0.0153	2.7141
Sentinel-2	2.5	CSR	64	40.9611	0.0074	1.5734
Sentinel-2	3.0	Bicubic	64	34.2978	0.0172	2.9960
Sentinel-2	3.0	CSR	64	38.9803	0.0091	1.8357
Sentinel-2	4.0	Bicubic	64	32.8090	0.0207	3.4275
Sentinel-2	4.0	CSR	64	38.8796	0.0096	1.8774
VENuS	1.0	Bicubic	36	52.5000	0.0016	0.4130
VENuS	1.0	CSR	36	65.9863	0.0004	0.1249
VENuS	1.5	Bicubic	36	45.5296	0.0035	0.8347
VENuS	1.5	CSR	36	54.9918	0.0012	0.4488
VENuS	2.0	Bicubic	36	42.8391	0.0048	1.0632
VENuS	2.0	CSR	36	50.7409	0.0020	0.6262
VENuS	2.5	Bicubic	36	41.3186	0.0057	1.2170
VENuS	2.5	CSR	36	47.7773	0.0028	0.7691
VENuS	3.0	Bicubic	36	40.2231	0.0065	1.3433
VENuS	3.0	CSR	36	44.9703	0.0039	1.0017
VENuS	4.0	Bicubic	36	38.3967	0.0081	1.6035
VENuS	4.0	CSR	36	42.4317	0.0050	1.1252

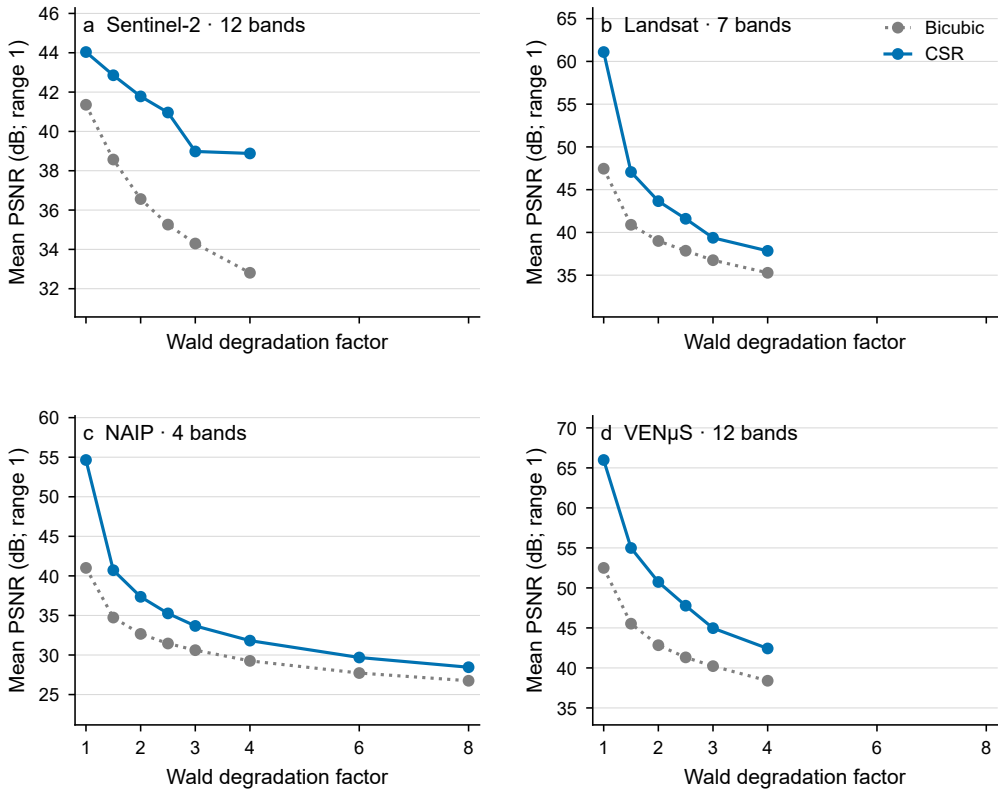


Figure A.12: Native reconstruction across MTF degradation factors, excluding zero targets. The pools exclude matching cache filenames and zero targets; NAIP values are scaled DN.

Table A.9: Common-budget adaptation with validation-only early stopping. Selected step zero is allowed when no trained checkpoint improves validation. Test references are evaluated after both checkpoint selections are frozen.

Initialization	Trained steps	Selected step	Validation cPSNR
CSR adapted	3250	1750	46.86
Random initialization	4750	3250	46.42

Table A.10: Adapted CSR minus each comparator in corrected PSNR on paired test scenes. Percentile 95% intervals from 10,000 bootstrap resamples of held-out scene/duplicate groups, seed zero; one adaptation training seed, no geographic-independence claim.

Band	Comparator	Scenes	Groups	Δ cPSNR	95% interval
RED	CSR pretrained	89	89	4.12	[3.71, 4.53]
RED	Bicubic	89	89	0.58	[0.49, 0.67]
RED	Random initialization	89	89	0.45	[0.39, 0.52]
NIR	CSR pretrained	85	85	3.75	[3.29, 4.24]
NIR	Bicubic	85	85	0.47	[0.38, 0.55]
NIR	Random initialization	85	85	0.37	[0.30, 0.43]

Table A.11: Complete metrics and finite-value counts for the matched BreizhSR adaptation cohort. Missing structural scores are excluded from their metric means.

Method	Metric	n	Mean
Bicubic	corrected psnr db	14823	33.870533
Bicubic	im metric	14820	0.072530
Bicubic	ha metric	14820	0.078368
Bicubic	om metric	14820	0.849102
Bicubic	reflectance	14823	0.004840
Bicubic	spectral	14813	0.983624
Bicubic	spatial	14823	0.003670
Bicubic	synthesis	14823	0.006393
Bicubic	unaligned psnr db	14823	32.350449
Bicubic	unaligned ssim	14823	0.804539
Bicubic	unaligned mae	14823	0.020889
Bicubic	unaligned rmse	14823	0.033186
Bicubic	unaligned sam deg	14819	7.302956
Bicubic	unaligned ergas	14815	15.247958
Lanczos	corrected psnr db	14823	33.870634
Lanczos	im metric	14820	0.072805
Lanczos	ha metric	14820	0.078932
Lanczos	om metric	14820	0.848263
Lanczos	reflectance	14823	0.004788
Lanczos	spectral	14813	0.959031
Lanczos	spatial	14823	0.003315
Lanczos	synthesis	14823	0.006474
Lanczos	unaligned psnr db	14823	32.338257
Lanczos	unaligned ssim	14823	0.803928
Lanczos	unaligned mae	14823	0.020900
Lanczos	unaligned rmse	14823	0.033216
Lanczos	unaligned sam deg	14820	7.314052
Lanczos	unaligned ergas	14815	15.266647
CSR pretrained	corrected psnr db	14823	31.504873
CSR pretrained	im metric	14820	0.150705
CSR pretrained	ha metric	14820	0.258288
CSR pretrained	om metric	14820	0.591007

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Table A.11: continued

Method	Metric	n	Mean
CSR pretrained	reflectance	14823	0.005500
CSR pretrained	spectral	14813	1.603385
CSR pretrained	spatial	14823	0.003313
CSR pretrained	synthesis	14823	0.011552
CSR pretrained	unaligned psnr db	14823	30.178357
CSR pretrained	unaligned ssim	14823	0.731492
CSR pretrained	unaligned mae	14823	0.024437
CSR pretrained	unaligned rmse	14823	0.039793
CSR pretrained	unaligned sam deg	14820	10.426366
CSR pretrained	unaligned ergas	14815	19.521142
CSR adapted	corrected psnr db	14823	34.327887
CSR adapted	im metric	14820	0.196138
CSR adapted	ha metric	14820	0.154798
CSR adapted	om metric	14820	0.649064
CSR adapted	reflectance	14823	0.008217
CSR adapted	spectral	14813	2.326744
CSR adapted	spatial	14822	0.655500
CSR adapted	synthesis	14823	0.010142
CSR adapted	unaligned psnr db	14823	32.677074
CSR adapted	unaligned ssim	14823	0.823584
CSR adapted	unaligned mae	14823	0.020126
CSR adapted	unaligned rmse	14823	0.032115
CSR adapted	unaligned sam deg	14820	7.078706
CSR adapted	unaligned ergas	14815	14.826479
Random init.	corrected psnr db	14823	33.921241
Random init.	im metric	14820	0.078754
Random init.	ha metric	14820	0.083300
Random init.	om metric	14820	0.837945
Random init.	reflectance	14823	0.005061
Random init.	spectral	14813	1.082539
Random init.	spatial	14823	0.088621
Random init.	synthesis	14823	0.006502
Random init.	unaligned psnr db	14823	32.426433
Random init.	unaligned ssim	14823	0.807004
Random init.	unaligned mae	14823	0.020774
Random init.	unaligned rmse	14823	0.032969
Random init.	unaligned sam deg	14820	7.290378
Random init.	unaligned ergas	14815	15.156653

Table A.12: Validation-selected BreizhSR checkpoints under the same maximum update budget. Selection uses corrected RGB PSNR on a fixed spatially balanced validation subset; test metrics enter only after both selections are frozen.

Initialization	Selected	Trained	Validation n	cPSNR
Pretrained	3250	4750	256	35.40
Random	1250	2750	256	34.95

Table A.13: Adapted CSR minus each matched baseline, with paired 95% spatial-block bootstrap intervals and metric-specific coverage. One adaptation seed.

Baseline	Metric	n	Blocks	Delta	Low	High
Bicubic	corrected psnr db	14823	259	0.4574	0.4030	0.5155
Bicubic	im metric	14820	259	0.1236	0.1139	0.1338
Bicubic	ha metric	14820	259	0.0764	0.0722	0.0807
Bicubic	om metric	14820	259	-0.2000	-0.2139	-0.1869
Bicubic	reflectance	14823	259	0.0034	0.0033	0.0035
Bicubic	spectral	14813	259	1.3431	1.2901	1.3979
Bicubic	spatial	14822	259	0.6518	0.6465	0.6570
Bicubic	synthesis	14823	259	0.0037	0.0034	0.0041
Bicubic	unaligned psnr db	14823	259	0.3266	0.2749	0.3752
Bicubic	unaligned ssim	14823	259	0.0190	0.0177	0.0203
Bicubic	unaligned mae	14823	259	-0.0008	-0.0008	-0.0007
Bicubic	unaligned rmse	14823	259	-0.0011	-0.0012	-0.0010
Bicubic	unaligned sam deg	14819	259	-0.2265	-0.2594	-0.1908
Bicubic	unaligned ergas	14815	259	-0.4215	-0.5203	-0.2992
Lanczos	corrected psnr db	14823	259	0.4573	0.4018	0.5165
Lanczos	im metric	14820	259	0.1233	0.1136	0.1336
Lanczos	ha metric	14820	259	0.0759	0.0718	0.0799
Lanczos	om metric	14820	259	-0.1992	-0.2128	-0.1862
Lanczos	reflectance	14823	259	0.0034	0.0033	0.0035
Lanczos	spectral	14813	259	1.3677	1.3137	1.4230
Lanczos	spatial	14822	259	0.6522	0.6468	0.6574
Lanczos	synthesis	14823	259	0.0037	0.0034	0.0040
Lanczos	unaligned psnr db	14823	259	0.3388	0.2870	0.3876
Lanczos	unaligned ssim	14823	259	0.0197	0.0183	0.0210
Lanczos	unaligned mae	14823	259	-0.0008	-0.0008	-0.0007
Lanczos	unaligned rmse	14823	259	-0.0011	-0.0012	-0.0010
Lanczos	unaligned sam deg	14820	259	-0.2353	-0.2688	-0.1978
Lanczos	unaligned ergas	14815	259	-0.4402	-0.5379	-0.3197
CSR pretrained	corrected psnr db	14823	259	2.8230	2.6841	2.9642
CSR pretrained	im metric	14820	259	0.0454	0.0331	0.0582
CSR pretrained	ha metric	14820	259	-0.1035	-0.1084	-0.0985
CSR pretrained	om metric	14820	259	0.0581	0.0434	0.0724
CSR pretrained	reflectance	14823	259	0.0027	0.0026	0.0028
CSR pretrained	spectral	14813	259	0.7234	0.6870	0.7624
CSR pretrained	spatial	14822	259	0.6522	0.6469	0.6573
CSR pretrained	synthesis	14823	259	-0.0014	-0.0017	-0.0012
CSR pretrained	unaligned psnr db	14823	259	2.4987	2.3884	2.6096
CSR pretrained	unaligned ssim	14823	259	0.0921	0.0889	0.0951
CSR pretrained	unaligned mae	14823	259	-0.0043	-0.0045	-0.0041
CSR pretrained	unaligned rmse	14823	259	-0.0077	-0.0080	-0.0074
CSR pretrained	unaligned sam deg	14820	259	-3.3477	-3.4783	-3.2218
CSR pretrained	unaligned ergas	14815	259	-4.6947	-5.3065	-4.2096
Random init.	corrected psnr db	14823	259	0.4066	0.3594	0.4569
Random init.	im metric	14820	259	0.1174	0.1079	0.1273
Random init.	ha metric	14820	259	0.0715	0.0679	0.0752
Random init.	om metric	14820	259	-0.1889	-0.2020	-0.1763
Random init.	reflectance	14823	259	0.0032	0.0030	0.0033
Random init.	spectral	14813	259	1.2442	1.1960	1.2941
Random init.	spatial	14822	259	0.5669	0.5619	0.5717
Random init.	synthesis	14823	259	0.0036	0.0033	0.0040

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Table A.13: continued

Baseline	Metric	n	Blocks	Delta	Low	High
Random init.	unaligned psnr db	14823	259	0.2506	0.2028	0.2953
Random init.	unaligned ssim	14823	259	0.0166	0.0154	0.0177
Random init.	unaligned mae	14823	259	-0.0006	-0.0007	-0.0006
Random init.	unaligned rmse	14823	259	-0.0009	-0.0009	-0.0008
Random init.	unaligned sam deg	14820	259	-0.2117	-0.2373	-0.1843
Random init.	unaligned ergas	14815	259	-0.3302	-0.4113	-0.2292

Appendix B. Guided Sentinel-2 reconstruction comparisons

Appendix B.1. Native-band L2A Wald comparison

We assess native-band fidelity in a complementary half-resolution Wald frame for all 178 source L2A scenes. Fine-band guidance is degraded from 10 to 20 m and coarse-band observations from 20 to 40 m. The target is the undegraded six-band native 20-m frame recovered by average pooling from the cached upsampled representation. CSR receives effective input GSDs reflecting the degradation; its recorded output is averaged back to the target grid. Bicubic, the official DSen2 weights and a community L2A implementation use the same guidance, coarse observations and target. This MTF-based diagnostic complements the Gaussian-degraded native products at the author-defined test footprints.

Table B.14: Matched L2A native-target Wald diagnostic on all source scenes. This uses synthetic MTF degradation and average-pool recovery of native coarse-band targets from an upsampled cache; its MTF and target construction differ from the guided native-product comparison. The L2A port denotes a community implementation.

Method	n	PSNR	RMSE	SAM	SSIM
Bicubic	178	36.36	0.0164	1.6344	0.8894
DSen2 author weights	178	39.25	0.0118	1.3482	0.9446
DSen2 L2A port	178	39.12	0.0120	1.3260	0.9430
CSR	178	44.21	0.0066	0.8290	0.9847

CSR reaches 44.21 dB in the matched native-target L2A diagnostic, compared with 39.25 dB for the author-released DSen2 weights and 39.12 dB for the community L2A implementation. Lower RMSE and SAM accompany this fidelity advantage (Table B.14).

Appendix B.2. Direct comparison on DSen2's guided task

We use the official DSen2 implementation and released weights (Lanaras et al., 2018), together with same-acquisition L2A products for the three released quantitative examples. The two configurations reconstruct six 20-m bands using four 10-m guidance bands, or two 60-m bands using the supplied finer bands. CSR receives the same guidance and coarse observations with their effective physical descriptors. Each test region is located by exact pixel matching between the released L1C reference band and its original native product; the resulting georeferenced footprint selects the L2A observations from the same platform, date and tile. This preserves the author-defined test locations while evaluating CSR on its declared Sentinel-2 processing level.

Native L2A bands provide the undegraded reference. The author's Gaussian blur and local pixel averaging reproduce the reduced-resolution inputs at factors two and six, followed by the official interpolation, overlapping patches and recomposition. DSen2 retains its released weights and DN/2000 network normalization; CSR receives the reflectance-scale DN/10000 convention. Predictions are scored against identical native references in common units. No model is fitted to these scenes. The original released L1C evaluation remains in the supplementary comparison to expose the effect of processing level under unchanged weights.

All three quantitative examples match the authors' test-tile list. The official PyTorch weight adapter is checked against TensorFlow on real, constant and zero inputs, with tolerance 10^{-4} in normalized output units. Reduced-precision CUDA matrix modes are disabled. Source products, native crop coordinates, band-array hashes, degraded input hashes and weight identities are retained with the measurements.

Table B.15: Guided reconstruction on same-acquisition L2A products for the three released DSen2 test footprints. Original native L1C pixel identity fixes each footprint; the author degradation, interpolation and tiling are reproduced on native L2A arrays. Identical guidance, coarse inputs and full-reference pixels are scored for all methods.

Tile	Factor	Method	PSNR	SSIM	MAE	SAM
T34HCH	6	Bicubic	30.64	0.7694	0.0169	2.26
T34HCH	6	DSen2	35.06	0.9242	0.0111	2.86
T34HCH	6	CSR	32.64	0.9349	0.0120	1.67
T43WFQ	2	Bicubic	34.81	0.9443	0.0108	2.25
T43WFQ	2	DSen2	47.33	0.9947	0.0029	1.45
T43WFQ	2	CSR	31.65	0.9276	0.0155	6.14
T18TWL	2	Bicubic	31.75	0.8538	0.0172	4.13
T18TWL	2	DSen2	41.08	0.9842	0.0060	3.21
T18TWL	2	CSR	26.26	0.8143	0.0295	7.62

The same-acquisition L2A comparison in Table B.15 retains the released test footprints and guided reconstruction task while aligning the processing level with CSR's training inputs. DSen2 leads PSNR in all 3 examples. In the factor-six case, CSR achieves higher SSIM and lower spectral angle, recovering a favorable structural and spectral balance with the shared reconstruction model. Figure B.13 shows the reconstructed target bands under a shared display stretch. The two L2A evaluations use different degradation and reference constructions: the native-target diagnostic averages the recorded finer-grid CSR prediction back to the cached native-band frame, while the released-footprint comparison scores predictions directly against undegraded native L2A bands under the author's Gaussian protocol. Their results place CSR's native-band fidelity alongside DSen2's strength on its specialized guided task.

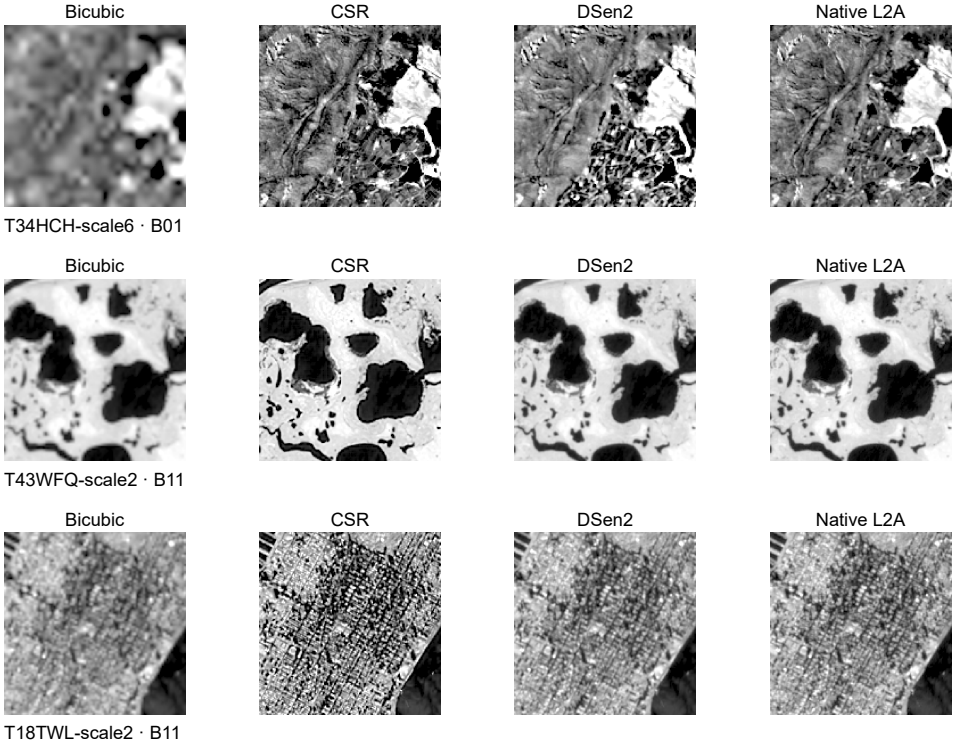


Figure B.13: Same-acquisition L2A reconstruction at the three released DSen2 quantitative footprints. The fixed B11 target is shown at factor two and B01 at factor six. Central crops and one input-derived stretch per example are shared across methods; native L2A bands provide the reference.

The factor-six B01 crop shows recovery of the terrain patterns seen in the native reference. The factor-two B11 crops also expose a principal reconstruction error mode: amplified local contrast around dark-water boundaries and built-up textures. These direct native-reference examples complement the cross-sensor error maps in Section 4.5. Complete target-band scores appear in Table B.16. The original released L1C measurements and their timing comparison appear in Tables B.17 and B.18.

Table B.16: Complete bandwise scores on the matched L2A acquisitions. Range-one PSNR and reflectance-scale MAE.

Tile	Factor	Method	Band	PSNR	SSIM	MAE
T34HCH	6	DSen2	all	35.06	0.9242	0.0111
T34HCH	6	DSen2	B01	35.03	0.9044	0.0113
T34HCH	6	DSen2	B09	35.09	0.9440	0.0109
T34HCH	6	CSR	all	32.64	0.9349	0.0120
T34HCH	6	CSR	B01	41.33	0.9741	0.0047
T34HCH	6	CSR	B09	29.93	0.8956	0.0192
T34HCH	6	Bicubic	all	30.64	0.7694	0.0169
T34HCH	6	Bicubic	B01	37.51	0.8975	0.0083
T34HCH	6	Bicubic	B09	28.11	0.6413	0.0254
T43WFQ	2	DSen2	all	47.33	0.9947	0.0029
T43WFQ	2	DSen2	B05	50.78	0.9944	0.0020
T43WFQ	2	DSen2	B06	47.03	0.9954	0.0032
T43WFQ	2	DSen2	B07	45.71	0.9953	0.0036
T43WFQ	2	DSen2	B8A	45.58	0.9956	0.0037
T43WFQ	2	DSen2	B11	47.46	0.9931	0.0030
T43WFQ	2	DSen2	B12	49.82	0.9946	0.0022
T43WFQ	2	CSR	all	31.65	0.9276	0.0155
T43WFQ	2	CSR	B05	39.63	0.9546	0.0065
T43WFQ	2	CSR	B06	30.67	0.9122	0.0190
T43WFQ	2	CSR	B07	29.14	0.9079	0.0226
T43WFQ	2	CSR	B8A	28.83	0.9101	0.0232
T43WFQ	2	CSR	B11	33.87	0.9351	0.0131
T43WFQ	2	CSR	B12	37.02	0.9457	0.0086
T43WFQ	2	Bicubic	all	34.81	0.9443	0.0108
T43WFQ	2	Bicubic	B05	43.72	0.9750	0.0042
T43WFQ	2	Bicubic	B06	33.79	0.9267	0.0134
T43WFQ	2	Bicubic	B07	32.15	0.9178	0.0162
T43WFQ	2	Bicubic	B8A	31.79	0.9201	0.0168
T43WFQ	2	Bicubic	B11	37.57	0.9587	0.0086
T43WFQ	2	Bicubic	B12	41.08	0.9677	0.0056
T18TWL	2	DSen2	all	41.08	0.9842	0.0060
T18TWL	2	DSen2	B05	39.67	0.9780	0.0069
T18TWL	2	DSen2	B06	41.27	0.9851	0.0059
T18TWL	2	DSen2	B07	41.23	0.9860	0.0060
T18TWL	2	DSen2	B8A	41.41	0.9868	0.0059
T18TWL	2	DSen2	B11	42.02	0.9855	0.0054
T18TWL	2	DSen2	B12	41.25	0.9837	0.0057
T18TWL	2	CSR	all	26.26	0.8143	0.0295
T18TWL	2	CSR	B05	25.40	0.7959	0.0306
T18TWL	2	CSR	B06	25.77	0.8027	0.0313
T18TWL	2	CSR	B07	26.09	0.8163	0.0310
T18TWL	2	CSR	B8A	26.50	0.8286	0.0300
T18TWL	2	CSR	B11	27.21	0.8253	0.0269
T18TWL	2	CSR	B12	26.87	0.8172	0.0268
T18TWL	2	Bicubic	all	31.75	0.8538	0.0172
T18TWL	2	Bicubic	B05	31.27	0.8364	0.0176
T18TWL	2	Bicubic	B06	31.29	0.8350	0.0184
T18TWL	2	Bicubic	B07	31.03	0.8342	0.0192
T18TWL	2	Bicubic	B8A	31.08	0.8403	0.0192
T18TWL	2	Bicubic	B11	33.20	0.8870	0.0146
T18TWL	2	Bicubic	B12	33.28	0.8902	0.0141

Appendix B.3. Released L1C examples and computational cost

Table B.17: Matched guided reconstruction on all author-released examples with supplied ground truth. Same fine-band guidance and coarse inputs; unaligned full-reference scoring. The cohort comprises the three released quantitative examples.

Tile	Factor	Method	PSNR	SSIM	MAE	SAM
T34HCH	6	Bicubic	37.78	0.9032	0.0081	2.36
T34HCH	6	DSen2	53.80	0.9973	0.0014	0.42
T34HCH	6	CSR	38.03	0.9336	0.0084	1.94
T43WFQ	2	Bicubic	36.04	0.9514	0.0092	1.79
T43WFQ	2	DSen2	50.11	0.9972	0.0021	0.93
T43WFQ	2	CSR	32.57	0.9336	0.0136	4.36
T18TWL	2	Bicubic	33.41	0.8700	0.0143	3.09
T18TWL	2	DSen2	43.85	0.9890	0.0043	2.20
T18TWL	2	CSR	28.03	0.8251	0.0246	5.40

The released L1C arrays retain their original processing and degradation conventions. DSen2 leads CSR on these examples, documenting the effect of applying CSR outside its declared L2A preprocessing under this guided task.

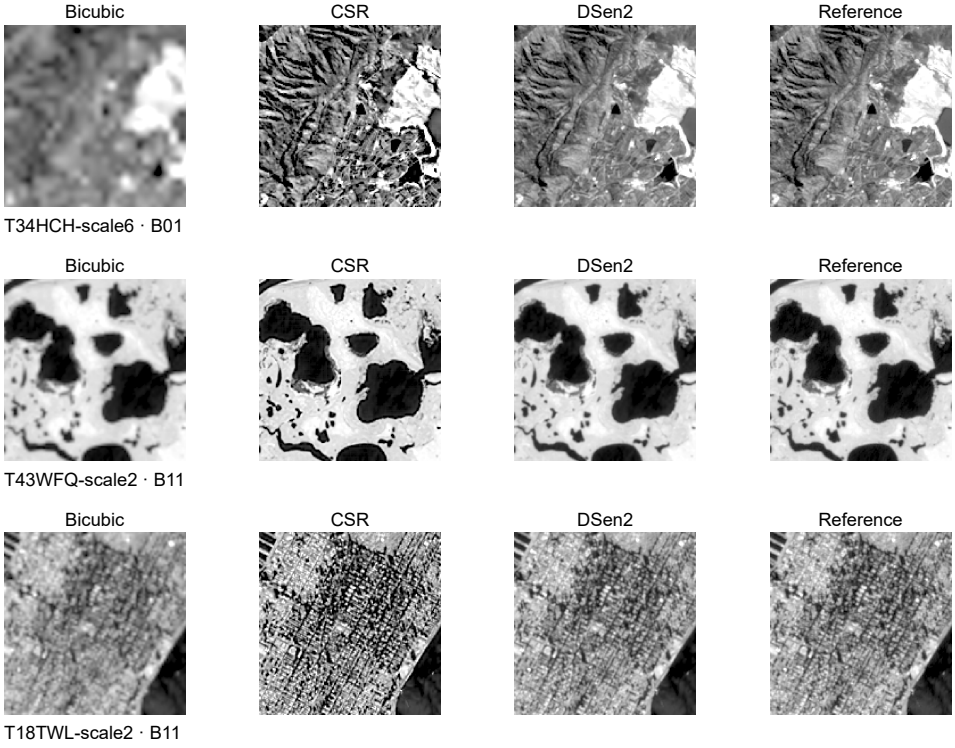


Figure B.14: Original released L1C guided examples with supplied reference arrays, shown with the same fixed target bands and display procedure as the L2A comparison in Figure B.13.

Table B.18: Mean GPU time per author-protocol patch on the same RTX A5000, with untimed warm-up. Both networks are evaluated in PyTorch after adapter validation. CSR returns all supplied bands; DSen2 returns target coarse bands. Patch sizes and band counts vary by task; this is a workload comparison.

Factor	Method	Parameters	Returned bands	Seconds/patch
2	DSen2	1,789,574	6	0.0038
2	CSR	24,572,065	10	1.0888
6	DSen2	1,787,266	2	0.0078
6	CSR	24,572,065	12	3.1260

DSen2's specialized architecture requires fewer parameters and less time

on these released-array workloads. The timing measurements reflect the output-band scope and patch configurations of each model.

Table B.19: Complete target-band scores for the released DSen2 examples. Range-one PSNR; physical reflectance MAE; SAM in degrees.

Tile	Factor	Method	Band	PSNR	SSIM	MAE
T34HCH	6	DSen2	all	53.80	0.9973	0.0014
T34HCH	6	DSen2	B01	57.53	0.9985	0.0009
T34HCH	6	DSen2	B09	51.83	0.9960	0.0018
T34HCH	6	CSR	all	38.03	0.9336	0.0084
T34HCH	6	CSR	B01	39.90	0.9391	0.0070
T34HCH	6	CSR	B09	36.73	0.9282	0.0098
T34HCH	6	Bicubic	all	37.78	0.9032	0.0081
T34HCH	6	Bicubic	B01	42.32	0.9563	0.0049
T34HCH	6	Bicubic	B09	35.61	0.8501	0.0113
T43WFQ	2	DSen2	all	50.11	0.9972	0.0021
T43WFQ	2	DSen2	B05	57.75	0.9986	0.0010
T43WFQ	2	DSen2	B06	49.40	0.9965	0.0024
T43WFQ	2	DSen2	B07	47.65	0.9962	0.0029
T43WFQ	2	DSen2	B8A	47.42	0.9965	0.0030
T43WFQ	2	DSen2	B11	51.56	0.9972	0.0019
T43WFQ	2	DSen2	B12	55.05	0.9981	0.0012
T43WFQ	2	CSR	all	32.57	0.9336	0.0136
T43WFQ	2	CSR	B05	42.41	0.9693	0.0046
T43WFQ	2	CSR	B06	32.18	0.9184	0.0158
T43WFQ	2	CSR	B07	29.97	0.9089	0.0204
T43WFQ	2	CSR	B8A	29.16	0.9085	0.0224
T43WFQ	2	CSR	B11	34.89	0.9396	0.0116
T43WFQ	2	CSR	B12	39.32	0.9568	0.0066
T43WFQ	2	Bicubic	all	36.04	0.9514	0.0092
T43WFQ	2	Bicubic	B05	46.37	0.9836	0.0031
T43WFQ	2	Bicubic	B06	35.69	0.9373	0.0108
T43WFQ	2	Bicubic	B07	33.40	0.9247	0.0140
T43WFQ	2	Bicubic	B8A	32.51	0.9239	0.0155
T43WFQ	2	Bicubic	B11	38.68	0.9627	0.0076
T43WFQ	2	Bicubic	B12	43.41	0.9759	0.0043
T18TWL	2	DSen2	all	43.85	0.9890	0.0043
T18TWL	2	DSen2	B05	43.77	0.9874	0.0042
T18TWL	2	DSen2	B06	43.65	0.9879	0.0045
T18TWL	2	DSen2	B07	43.10	0.9881	0.0049
T18TWL	2	DSen2	B8A	42.98	0.9889	0.0050
T18TWL	2	DSen2	B11	45.07	0.9914	0.0038
T18TWL	2	DSen2	B12	44.95	0.9901	0.0037
T18TWL	2	CSR	all	28.03	0.8251	0.0246
T18TWL	2	CSR	B05	27.94	0.8119	0.0233
T18TWL	2	CSR	B06	27.72	0.8127	0.0257
T18TWL	2	CSR	B07	27.44	0.8208	0.0273
T18TWL	2	CSR	B8A	27.27	0.8281	0.0282
T18TWL	2	CSR	B11	28.68	0.8368	0.0231
T18TWL	2	CSR	B12	29.56	0.8403	0.0200
T18TWL	2	Bicubic	all	33.41	0.8700	0.0143
T18TWL	2	Bicubic	B05	33.76	0.8632	0.0133
T18TWL	2	Bicubic	B06	33.21	0.8551	0.0149

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Table B.19: continued

Tile	Factor	Method	Band	PSNR	SSIM	MAE
T18TWL	2	Bicubic	B07	32.44	0.8482	0.0165
T18TWL	2	Bicubic	B8A	31.99	0.8490	0.0174
T18TWL	2	Bicubic	B11	34.47	0.8967	0.0128
T18TWL	2	Bicubic	B12	35.57	0.9078	0.0109

Appendix C. Inference sensitivities

Native diagnostic pools remove filenames shared with the identified mission caches and wholly zero targets, and cover integer and noninteger degradation factors. Paired sensitivities use 128 native patch/scale cases across the training missions and 7 inference variants with unchanged weights. Each variant receives identical inputs and targets; low-frequency agreement and degradation-back MAE use the prespecified baseline MTF. The measurements characterize inference behavior; retrained component controls remain a separate experiment design.

Table C.20: Same-weight inference sensitivities. Identical seeded patches and central targets at each scale; no retraining. Degrade-back MAE uses the prespecified baseline MTF for all variants. The cases describe native-band reconstruction within the sampled diagnostic pools.

Mission	Scale	Variant	N	PSNR	MAE	Back MAE
Sentinel-2	2	CSR	16	42.76	0.0060	0.00149
Sentinel-2	2	Before composition	16	27.14	0.0449	0.04297
Sentinel-2	2	No jitter	16	42.76	0.0060	0.00149
Sentinel-2	2	Four neighbours	16	42.66	0.0061	0.00149
Sentinel-2	2	Permuted band metadata	16	42.24	0.0064	0.00143
Sentinel-2	2	MTF lower	16	42.79	0.0060	0.00181
Sentinel-2	2	MTF higher	16	42.62	0.0061	0.00135
Sentinel-2	4	CSR	16	39.52	0.0086	0.00218
Sentinel-2	4	Before composition	16	25.11	0.0512	0.04828
Sentinel-2	4	No jitter	16	39.52	0.0086	0.00218
Sentinel-2	4	Four neighbours	16	39.48	0.0086	0.00219
Sentinel-2	4	Permuted band metadata	16	38.90	0.0092	0.00216
Sentinel-2	4	MTF lower	16	39.54	0.0085	0.00265
Sentinel-2	4	MTF higher	16	39.32	0.0088	0.00197
Landsat-8/9	2	CSR	16	42.90	0.0046	0.00036
Landsat-8/9	2	Before composition	16	31.01	0.0249	0.02322
Landsat-8/9	2	No jitter	16	42.90	0.0046	0.00036
Landsat-8/9	2	Four neighbours	16	42.75	0.0047	0.00040
Landsat-8/9	2	Permuted band metadata	16	42.63	0.0047	0.00045
Landsat-8/9	2	MTF lower	16	43.00	0.0045	0.00024
Landsat-8/9	2	MTF higher	16	42.74	0.0047	0.00053
Landsat-8/9	4	CSR	16	37.73	0.0082	0.00058
Landsat-8/9	4	Before composition	16	30.66	0.0255	0.02288
Landsat-8/9	4	No jitter	16	37.73	0.0082	0.00058

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Table C.20: continued

Mission	Scale	Variant	N	PSNR	MAE	Back MAE
Landsat-8/9	4	Four neighbours	16	37.67	0.0082	0.00059
Landsat-8/9	4	Permuted band metadata	16	37.60	0.0083	0.00058
Landsat-8/9	4	MTF lower	16	37.78	0.0081	0.00042
Landsat-8/9	4	MTF higher	16	37.66	0.0083	0.00075
NAIP	2	CSR	16	36.50	0.0132	0.00109
NAIP	2	Before composition	16	26.36	0.0468	0.03953
NAIP	2	No jitter	16	36.50	0.0132	0.00109
NAIP	2	Four neighbours	16	36.26	0.0135	0.00126
NAIP	2	Permuted band metadata	16	36.35	0.0133	0.00101
NAIP	2	MTF lower	16	36.62	0.0130	0.00079
NAIP	2	MTF higher	16	36.30	0.0135	0.00153
NAIP	4	CSR	16	30.59	0.0244	0.00160
NAIP	4	Before composition	16	25.49	0.0497	0.03801
NAIP	4	No jitter	16	30.59	0.0244	0.00160
NAIP	4	Four neighbours	16	30.53	0.0245	0.00169
NAIP	4	Permuted band metadata	16	30.45	0.0247	0.00163
NAIP	4	MTF lower	16	30.64	0.0241	0.00116
NAIP	4	MTF higher	16	30.51	0.0247	0.00210
VENuS	2	CSR	16	50.79	0.0020	0.00030
VENuS	2	Before composition	16	31.80	0.0224	0.02145
VENuS	2	No jitter	16	50.79	0.0020	0.00030
VENuS	2	Four neighbours	16	50.30	0.0021	0.00032
VENuS	2	Permuted band metadata	16	50.46	0.0021	0.00033
VENuS	2	MTF lower	16	51.01	0.0019	0.00021
VENuS	2	MTF higher	16	50.41	0.0021	0.00042
VENuS	4	CSR	16	42.70	0.0050	0.00055
VENuS	4	Before composition	16	32.04	0.0220	0.02050
VENuS	4	No jitter	16	42.70	0.0050	0.00055
VENuS	4	Four neighbours	16	42.68	0.0050	0.00055
VENuS	4	Permuted band metadata	16	42.51	0.0051	0.00052
VENuS	4	MTF lower	16	42.80	0.0049	0.00041
VENuS	4	MTF higher	16	42.57	0.0050	0.00068

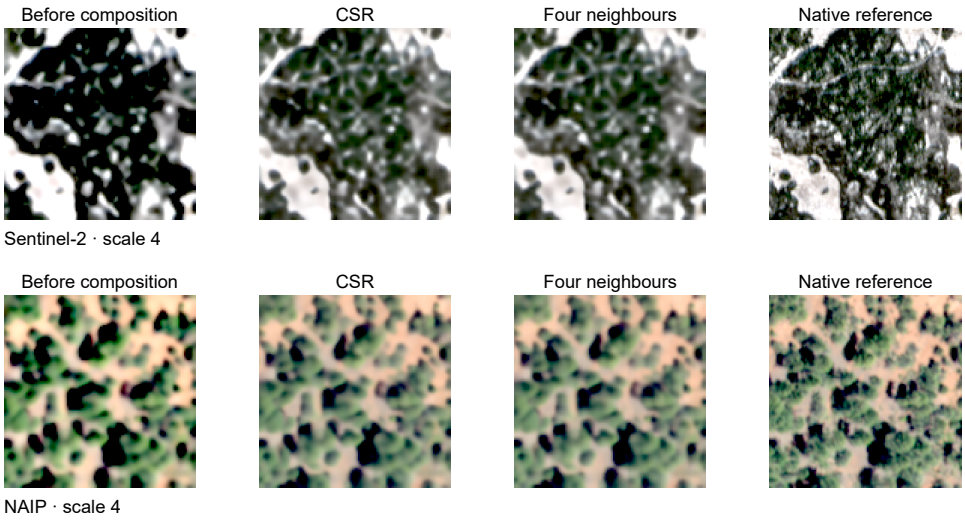


Figure C.15: Fixed first selected diagnostic patches at scale four. Before-composition output, the trained CSR query rule, same-weight four-neighbour diagnostic queries and native reference share an input-derived stretch. The variants share the trained parameters and isolate changes to inference.

Table C.21: Prespecified first labelled scenes per band; MTF sensitivity with the same fixed checkpoint and inputs. No reference-based MTF selection.

Band	Method	MTF	Scenes	cPSNR	Direct PSNR
RED	Bicubic	–	16	47.98	42.92
RED	Lanczos	–	16	48.08	42.93
RED	CSR	0.1	16	43.47	39.53
RED	CSR	0.2	16	44.40	40.07
RED	CSR	0.3	16	45.22	40.56
NIR	Bicubic	–	16	44.52	39.34
NIR	Lanczos	–	16	44.60	39.34
NIR	CSR	0.1	16	39.54	35.83
NIR	CSR	0.2	16	40.43	36.38
NIR	CSR	0.3	16	41.23	36.86

Appendix D. LS2S2 supplementary diagnostics

The released preview contains one AOI, 31TCJ_12. Its expansion considers 28 valid Landsat dates at nine fixed crop centres and six degradation

factors. It scores 278 date/crop/scale cells and excludes 1234 cells below 98% clear support or outside the grid. Seven reflective Landsat bands are evaluated. The cells are correlated observations within one AOI; they describe single-image reconstruction rather than temporal fusion.

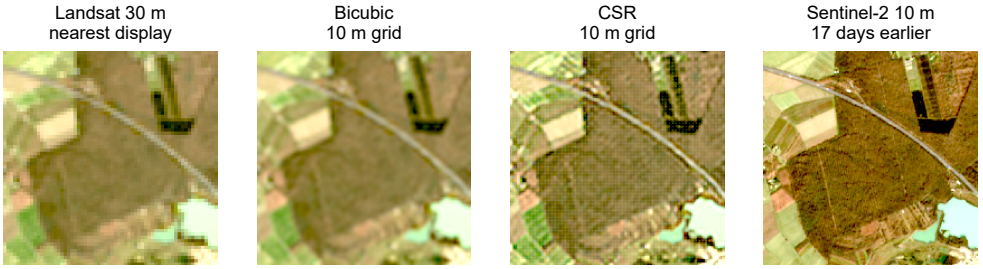


Figure D.16: Landsat input, bicubic, CSR and different-date Sentinel-2 context for the recorded preview case, with a shared input-derived stretch. Sentinel-2 is contextual and is not pixel ground truth.

Table D.22: Full downloaded LS2S2 preview: every valid acquisition is considered at nine fixed crop centres. Cells require at least 98% cloud/nodata-free target support. Counts are crops and dates, not independent AOIs. Seven reflective bands only; SAM is reported only for joint spectra.

Scale	Method	Band	Crops	Dates	PSNR	MAE	RMSE	SAM
1.0	Bicubic	B1	93	22	52.9633	0.0015	0.0023	–
1.0	Bicubic	B2	93	22	51.2450	0.0019	0.0028	–
1.0	Bicubic	B3	93	22	48.8681	0.0025	0.0037	–
1.0	Bicubic	B4	93	22	45.8240	0.0036	0.0053	–
1.0	Bicubic	B5	93	22	42.0050	0.0057	0.0081	–
1.0	Bicubic	B6	93	22	42.9576	0.0051	0.0073	–
1.0	Bicubic	B7	93	22	43.7704	0.0047	0.0066	–
1.0	Bicubic	all	93	22	45.1821	0.0036	0.0056	1.0385
1.0	CSR	B1	93	22	71.0707	0.0002	0.0003	–
1.0	CSR	B2	93	22	69.5709	0.0002	0.0003	–
1.0	CSR	B3	93	22	67.2530	0.0003	0.0004	–
1.0	CSR	B4	93	22	64.4212	0.0004	0.0006	–
1.0	CSR	B5	93	22	60.3943	0.0007	0.0010	–
1.0	CSR	B6	93	22	61.3422	0.0006	0.0009	–
1.0	CSR	B7	93	22	62.8854	0.0005	0.0007	–
1.0	CSR	all	93	22	63.7456	0.0004	0.0007	0.1388
1.5	Bicubic	B1	78	20	46.2210	0.0033	0.0050	–
1.5	Bicubic	B2	78	20	44.4963	0.0041	0.0061	–
1.5	Bicubic	B3	78	20	41.9842	0.0055	0.0081	–
1.5	Bicubic	B4	78	20	38.9983	0.0079	0.0115	–

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Table D.22: continued

Scale	Method	Band	Crops	Dates	PSNR	MAE	RMSE	SAM
1.5	Bicubic	B5	78	20	35.3475	0.0123	0.0173	–
1.5	Bicubic	B6	78	20	36.3421	0.0110	0.0155	–
1.5	Bicubic	B7	78	20	37.0984	0.0101	0.0143	–
1.5	Bicubic	all	78	20	38.4967	0.0078	0.0120	2.2800
1.5	CSR	B1	78	20	55.8430	0.0011	0.0016	–
1.5	CSR	B2	78	20	54.2948	0.0013	0.0020	–
1.5	CSR	B3	78	20	51.9622	0.0017	0.0026	–
1.5	CSR	B4	78	20	49.2852	0.0024	0.0035	–
1.5	CSR	B5	78	20	44.5746	0.0043	0.0059	–
1.5	CSR	B6	78	20	46.0269	0.0036	0.0051	–
1.5	CSR	B7	78	20	47.0191	0.0031	0.0045	–
1.5	CSR	all	78	20	48.2105	0.0025	0.0039	0.7754
2.0	Bicubic	B1	45	17	44.3370	0.0042	0.0062	–
2.0	Bicubic	B2	45	17	42.6165	0.0052	0.0076	–
2.0	Bicubic	B3	45	17	40.1278	0.0069	0.0101	–
2.0	Bicubic	B4	45	17	37.1143	0.0101	0.0143	–
2.0	Bicubic	B5	45	17	33.1734	0.0160	0.0222	–
2.0	Bicubic	B6	45	17	34.1625	0.0143	0.0199	–
2.0	Bicubic	B7	45	17	35.0062	0.0131	0.0181	–
2.0	Bicubic	all	45	17	36.3836	0.0100	0.0153	3.0553
2.0	CSR	B1	45	17	51.1059	0.0019	0.0028	–
2.0	CSR	B2	45	17	49.6678	0.0022	0.0033	–
2.0	CSR	B3	45	17	47.3922	0.0029	0.0043	–
2.0	CSR	B4	45	17	44.6863	0.0041	0.0059	–
2.0	CSR	B5	45	17	39.7639	0.0074	0.0103	–
2.0	CSR	B6	45	17	41.1378	0.0062	0.0089	–
2.0	CSR	B7	45	17	42.1897	0.0055	0.0079	–
2.0	CSR	all	45	17	43.4075	0.0043	0.0068	1.3361
2.5	Bicubic	B1	54	16	43.2861	0.0048	0.0070	–
2.5	Bicubic	B2	54	16	41.5520	0.0059	0.0086	–
2.5	Bicubic	B3	54	16	38.9928	0.0080	0.0115	–
2.5	Bicubic	B4	54	16	35.9697	0.0116	0.0163	–
2.5	Bicubic	B5	54	16	31.9877	0.0184	0.0255	–
2.5	Bicubic	B6	54	16	33.0145	0.0165	0.0227	–
2.5	Bicubic	B7	54	16	33.8729	0.0150	0.0207	–
2.5	Bicubic	all	54	16	35.2322	0.0115	0.0175	3.4630
2.5	CSR	B1	54	16	48.3501	0.0026	0.0039	–
2.5	CSR	B2	54	16	46.8708	0.0031	0.0046	–
2.5	CSR	B3	54	16	44.5526	0.0041	0.0060	–
2.5	CSR	B4	54	16	41.7906	0.0057	0.0083	–
2.5	CSR	B5	54	16	37.1178	0.0101	0.0140	–
2.5	CSR	B6	54	16	38.4086	0.0086	0.0121	–
2.5	CSR	B7	54	16	39.3556	0.0076	0.0109	–
2.5	CSR	all	54	16	40.6501	0.0059	0.0093	1.7855
3.0	Bicubic	B1	2	2	43.7040	0.0044	0.0067	–
3.0	Bicubic	B2	2	2	42.0825	0.0054	0.0081	–
3.0	Bicubic	B3	2	2	39.2481	0.0076	0.0111	–
3.0	Bicubic	B4	2	2	36.6937	0.0106	0.0152	–
3.0	Bicubic	B5	2	2	32.0463	0.0182	0.0250	–
3.0	Bicubic	B6	2	2	33.2896	0.0155	0.0219	–
3.0	Bicubic	B7	2	2	34.7735	0.0135	0.0188	–
3.0	Bicubic	all	2	2	35.6110	0.0107	0.0167	3.2368
3.0	CSR	B1	2	2	46.9099	0.0031	0.0046	–

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Table D.22: continued

Scale	Method	Band	Crops	Dates	PSNR	MAE	RMSE	SAM
3.0	CSR	B2	2	2	45.7495	0.0035	0.0052	–
3.0	CSR	B3	2	2	43.4713	0.0046	0.0068	–
3.0	CSR	B4	2	2	41.1026	0.0062	0.0090	–
3.0	CSR	B5	2	2	35.8627	0.0117	0.0162	–
3.0	CSR	B6	2	2	37.4544	0.0095	0.0134	–
3.0	CSR	B7	2	2	38.8776	0.0080	0.0116	–
3.0	CSR	all	2	2	39.6666	0.0066	0.0104	2.0350
4.0	Bicubic	B1	6	6	41.3011	0.0062	0.0088	–
4.0	Bicubic	B2	6	6	39.5256	0.0076	0.0108	–
4.0	Bicubic	B3	6	6	36.8236	0.0104	0.0147	–
4.0	Bicubic	B4	6	6	33.8471	0.0151	0.0210	–
4.0	Bicubic	B5	6	6	29.2277	0.0255	0.0348	–
4.0	Bicubic	B6	6	6	31.0148	0.0210	0.0286	–
4.0	Bicubic	B7	6	6	31.9267	0.0192	0.0260	–
4.0	Bicubic	all	6	6	33.0099	0.0150	0.0227	4.5484
4.0	CSR	B1	6	6	44.4295	0.0041	0.0061	–
4.0	CSR	B2	6	6	42.8941	0.0049	0.0073	–
4.0	CSR	B3	6	6	40.3912	0.0065	0.0097	–
4.0	CSR	B4	6	6	37.5607	0.0093	0.0135	–
4.0	CSR	B5	6	6	32.4158	0.0173	0.0241	–
4.0	CSR	B6	6	6	34.4029	0.0137	0.0192	–
4.0	CSR	B7	6	6	35.4060	0.0122	0.0173	–
4.0	CSR	all	6	6	36.3999	0.0097	0.0152	2.9098

Declaration of generative AI and AI-assisted technologies

OpenAI Codex was used to assist with literature retrieval and source verification, manuscript drafting and editing, implementation of evaluation and visualization code, and LaTeX preparation. Numerical results are regenerated from recorded evaluation outputs, and displayed imagery derives from recorded observations and model predictions. The authors retain responsibility for reviewing the AI-assisted material, the scientific interpretations and the final approved manuscript.

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